



TEAMCENTER

Teamcenter Upgrade

Teamcenter 14.3

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Support Center: support.sw.siemens.com

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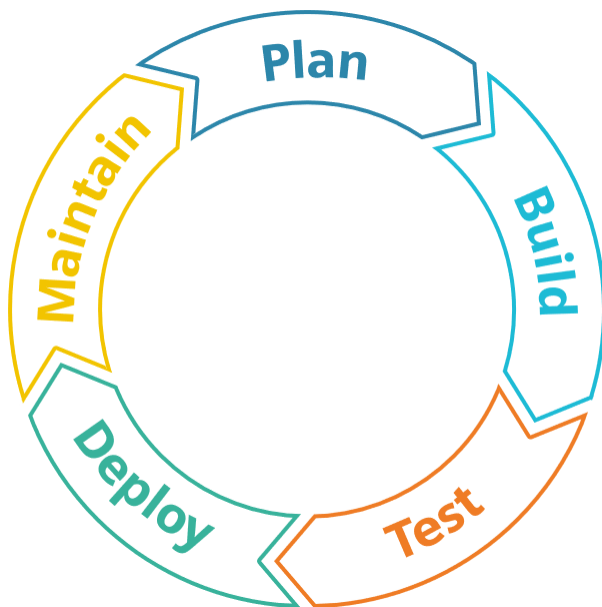
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1. Upgrading Teamcenter with Active Workspace

Upgrading Teamcenter with Active Workspace follows a similar five-phase process as installing the Teamcenter environment with Active Workspace.



Plan

Download software and documentation, and devise your upgrade strategy based on the contents of your Teamcenter environment.

Build

Update your infrastructure components, such as your database server, and create a test environment for upgrade.

Test

Upgrade a test system to validate your upgrade strategy and identify any potential issues.

Deploy

Deploy your test environment configuration to your production environment, rolling out the latest Teamcenter and Active Workspace changes to your users.

Maintain

Perform ongoing maintenance of your Teamcenter environment, and updating software with the latest patches.

When a new major version of Teamcenter is released, this cycle repeats as you begin planning the next upgrade.

If you do not use Active Workspace, but you use the Teamcenter *rich client*,¹ you can upgrade rich clients at the same time as you upgrade Teamcenter servers, or upgrade them after.

¹ The Teamcenter rich client is a Java-based desktop client that is an alternative to Active Workspace.

Where do I go from here?

You can begin with *Plan the Teamcenter upgrade*, or proceed to one of the following common upgrade tasks:

 Administrator	
Downloading software and documentation, learning about upgrading templates	See <i>Plan the Teamcenter upgrade</i> .
Upgrading database, updating licenses	See <i>Build the Teamcenter upgrade test environment</i> .
Upgrading a test environment	See <i>Test the Teamcenter upgrade</i> .
Updating Active Workspace	See <i>Update Active Workspace and microservices</i> .
Upgrading a rich client	See <i>Rich Client Installation on Windows</i> or <i>Rich Client Installation on Linux</i> .
Upgrading a production environment	See <i>Deploy the Teamcenter upgrade</i> .
Applying Teamcenter patches	See <i>Installing Teamcenter patches</i> .

Part I: Plan the Teamcenter upgrade



Plan your Teamcenter upgrade in consideration of all software components in your environment, including infrastructure components like database servers and license server, which may need upgrading before you perform the Teamcenter upgrade.

Plan to perform an upgrade on a test environment to identify and resolve potential issues before you upgrade production systems.

The *Teamcenter Deployment Reference Architecture*, available from the Teamcenter downloads page on Support Center, is an essential resource during upgrade. It provides information such as:

- Guidelines for copying a Teamcenter environment for upgrade testing.
- Detailed examples of Teamcenter and Active Workspace deployments.
- Sample configurations and scripts to use with Deployment Center.

Use the *Teamcenter Deployment Reference Architecture* to help plan your upgrade of Teamcenter with Active Workspace. Then proceed to building a test environment for upgrade.

2. Where to start

Get documentation

Teamcenter documentation is available from two sources:

- **Support Center**

Siemens Digital Industries Software's comprehensive support portal, which provides documentation for all Siemens software products and versions.

The *Siemens Secure Documentation Proxy* provides secure, easy access to documentation without a need to log on, through a personalized API key specific to your site.

- **Siemens Documentation Server**

Locally installed server that can host documentation for all your Siemens Digital Industries Software products. No Internet access is required. You can configure the server for single-machine or network-wide access.

The Siemens Secure Documentation Proxy and the Siemens Documentation Server are available on Support Center under **Products**→**Siemens Documentation Installer**→**Downloads**.

Choose how you want to access documentation, then download and install the required software:

Installing Siemens Secure Documentation Proxy	Installing Siemens Documentation Server
Install the Siemens Secure Documentation Proxy to access documentation on Support Center.	1. Install the Siemens Documentation Server. 2. Install documentation for your software products. Each <i>documentation kit</i> contains documentation content and an installation wizard that automatically installs documentation onto your Documentation Server. Documentation kits (docs-product-name-and-version-locale.zip) are available from the Downloads pages for each product on Support Center.

Complete installation instructions for the Secure Documentation Proxy and the Siemens Documentation Server are available on Support Center.

For a step-by-step orientation to Support Center, see Siemens Software [Support Center videos](#) on YouTube.

View documentation for Teamcenter and Active Workspace

View documentation on Support Center at the following URLs:

- Teamcenter 14.3:

<https://docs.sw.siemens.com/en-US/product/282219420/doc/PL20221031257051471.xid1899404/html/xid1899405>

- Active Workspace 6.3:

<https://docs.sw.siemens.com/en-US/product/282219420/doc/PL20221031255989072.xid1899337/html/xid1899339>

- Deployment Center 14.3:

https://docs.sw.siemens.com/en-US/product/282219420/doc/PL20221031256176867.deployment_center/html/xid1161106

Enable rich client access to documentation

When prompted in the Teamcenter installation tools, supply the appropriate documentation URL to the rich client:

`http://domain/en-US/product/282219420/doc/PL20221031257051471.xid1899404/html/xid1899405`

Replace *domain* with the source from which you access documentation:

- Support Center: **docs.sw.siemens.com**
- Siemens Documentation Server: *doc-server-host:doc-server-port*

Enable Active Workspace access to documentation

If you use Active Workspace documentation on Support Center, no further configuration is necessary. The Active Workspace **Help** button links to Support Center by default.¹

If you install the Siemens Documentation Server on your local network, configure the **Help** button to link to Active Workspace documentation on your local server. Perform the following steps after you install Active Workspace:

¹ The default URL to Active Workspace documentation on Support Center is <http://docs.sw.siemens.com/en-US/product/282219420/doc/PL20221031255989072.xid1899337/html/xid1899339>.

1. In Active Workspace, open Command Builder.
2. Find the **showHelp** action and set its **Navigate To** property to the URL to Active Workspace documentation on your local documentation server:

`http://host:port/en-US/product/282219420/doc/PL20221031255989072.xid1899337/html/xid1899339`

Replace *host* and *port* with the host name and port of your local documentation server.

3. Commit your UI Builder changes to your module to update the **Help** button link for your users.

Get Deployment Center

Deployment Center is a centralized web application for deploying software to Teamcenter environments. Using Deployment Center, you can create and manage multiple environments from a single location. This simplifies the processes of installing and updating software and can automate deployment.

The screenshot displays the Deployment Center web application. The top navigation bar includes the user 'dcadmin (dcadmin)' and the application name 'Deployment Center SIEMENS'. The main section is titled 'Environments' with a sub-section 'All Environments'. Below this, there's a card for 'Env_001' (Type: Development, Location:). A progress bar indicates the deployment process with five steps: '1 Software', '2 Options', '3 Applications', '4 Components', and '5 Deploy'. The 'Selected Components' table lists the following components and their status:

COMPONENT	OS	COMPLETE	STATUS
Active Workspace Client Builder		Start	⌚
Active Workspace Gateway		Start	⌚
Corporate Server		Start	⌚
Database Server		Start	🇩🇪
FSC		Start	⌚
Indexer		Start	⌚
Indexing Engine		Start	⌚
Licensing Server		Start	🇩🇪
Microservice Node		Start	⌚

The 'Corporate Server' component is selected, and its configuration details are shown on the right. The 'Machine' section includes 'Machine Name' (empty field) and 'OS' (Inx64). The 'General Settings' section includes 'Teamcenter Installation Path' (empty field) and 'Teamcenter Administrative User' (empty field). A 'Start Configuration' button is located at the bottom of the components table, and a 'Save Component Settings' button is located at the bottom of the configuration panel.

Download the latest version of Deployment Center from the Teamcenter downloads area on Support Center. Install Deployment Center as described in the *Deployment Center Guide*.

Deployment Center is an alternative installation tool to Teamcenter Environment Manager (TEM) for installing Teamcenter and Active Workspace. TEM is deprecated and will be discontinued in a future release.

Get software

Upgrading Teamcenter with Active Workspace requires software for Teamcenter, microservice framework, and Active Workspace.

Download Teamcenter software

From the Teamcenter downloads page on Support Center, download the Teamcenter 14 and 14.3 software kits for your platform:

Windows:

Tc14.0.0.0_wntx64.zip
Tc14.3.0.0_wntx64.zip

Linux:

Tc14.0.0.0_Inx64.zip
Tc14.3.0.0_Inx64.zip

If a patch to Teamcenter 14.3 is available, for example, Teamcenter 14.3.0.1, you can alternatively substitute that kit for the Teamcenter 14.3 kit. Patch kit files are named **Tc14.3.0_patch_number_platform.zip**.

Download microservice framework software

From the Teamcenter downloads page, select **Additional Downloads**→**Microservice Framework**, and then download the microservice framework 6.3 software kit for your platform type. This kit is required for Active Workspace installation:

Windows:

TcMicroserviceFramework6.3.0_wntx64.zip

Linux:

TcMicroserviceFramework6.3.0_Inx64.zip

If a later microservice framework software patch kit is available, for example, version 6.3.1, you can download that kit instead.

Download Active Workspace software

From the Teamcenter downloads page on Support Center, download the Active Workspace 6.3 software kit for your version of Teamcenter and for your platform type:

Windows:

Tc14.x.0.0_ActiveWorkspace6.3.0_wntx64.zip

Linux:

Tc14.x.0.0_ActiveWorkspace6.3.0_lnx64.zip

If a later Active Workspace software patch kit is available, for example, version 6.3.1, you can download that kit instead.

Note:

Active Workspace 6.3 supports multiple versions of Teamcenter. Take care to download the correct software kit for Teamcenter 14.x.

Stage the software kits

Deployment Center

1. Expand the software kits for Teamcenter, microservice framework, and Active Workspace. Copy the unzipped directories to the *software* subdirectory in one of your registered repository locations.
2. Log on to Deployment Center, and click **SOFTWARE REPOSITORIES**.

The **Software Repositories** page opens the **Active Media** tab of the repository and displays the **Software Media** table.

3. Verify that the added software appears in the list of available software. The list may take up to five minutes to update.

If the software does not appear in the **Software Repositories** page, inspect the repository log files for repository scanning issues or software file problems. The repository log files are in the **webserver\reptool\logs** directory on the Deployment Center server.

Teamcenter Environment Manager (TEM)

Expand the software kits for Teamcenter, microservice framework, and Active Workspace to separate directories that are accessible to the machines on which you plan to install.

Can I place the software in a remote location?

You can place software kits on a non-local drive, with the following considerations.

Linux systems:

Deployment Center

Your primary repository in Deployment Center must be a local path. However, you can specify additional repository locations, and these may be UNC paths or local file system paths. Mapped drives are not supported for any software repositories in Deployment Center. For more information, see the *Deployment Center Guide*.

Teamcenter Environment Manager (TEM)

Windows systems:

TEM cannot install software from UNC paths, for example, `\\mediaserver\tcmedia`. If the software kits are located on a remote host, map a drive to each software location using the **net use** command.

Open an administrator command prompt and type the **net use** command:

```
net use drive-letter: UNC-path
```

For example:

```
net use z: \\mediaserver\tcmedia
```

Linux systems:

If you mount software kits on a remote NFS server, you must launch Teamcenter Environment Manager on the local server node.

System requirements

For versions of system software and hardware certified for running Teamcenter on your platform, see the Hardware and Software Certifications knowledge base article on Support Center:

<https://support.sw.siemens.com>

Some software requirements differ for non-English locales. When viewing certified versions, make sure you note any exceptions for your locale.

Hardware requirements for a Teamcenter deployment vary depending on considerations such as whether your deployment contains:

- A single host or multiple hosts
- Rich client, Active Workspace, or both
- Additional components such as Dispatcher Server on separate hosts

The *Teamcenter Server Hardware Overview* contains hardware recommendations based on these and other variables. This document is available from the **Support White Papers** *version* page in the Teamcenter downloads area on Support Center.²

Platforms

Determine from the following table which Teamcenter 14.3 components are supported on your operating system. Check marks (√) indicate components supported on the given operating system.

Operating system	Corporate server	Web tier	Active Workspace	Rich Client	Business Modeler IDE client	TCCS
Microsoft Windows (desktop platforms)			√	√	√	√
Microsoft Windows Server	√	√			√	
SUSE Linux	√	√	√	√	√	√
Red Hat Linux	√	√	√	√	√	√

Microsoft Windows

- On Windows platforms, disable Windows User Account Control (UAC) before you install Teamcenter. This option is available in the **Control Panel**→**User Accounts** dialog box.

Windows UAC can interfere with Teamcenter installation programs. Siemens Digital Industries Software recommends turning off UAC for administrative users only.

For more information, see Microsoft Windows documentation.

- Disable the Windows TCP scaling feature. Open a command prompt and enter the following command:

```
netsh interface tcp set global autotuninglevel=disabled
```

Siemens Digital Industries Software recommends setting this parameter before installing Teamcenter because most client network infrastructures use one or more switches or routers.

² The *Teamcenter Server Hardware Overview* is named **Teamcenter_Hardware_Overview-version.pdf**. The latest version of this document may be in an earlier **Support White Papers** *version* page.

By default, Windows enables TCP window scaling, but some routers do not support this feature. This can cause installation failures that are difficult to diagnose and correct.

For more information, see Microsoft Windows documentation.

- If you use a nonnative language operating system version of Windows, you must install and enable the Multilingual User Interface (MUI) pack to ensure the language font is displayed properly.
 1. Download and install the MUI pack for Windows from Microsoft.
 2. Open the **Regional and Language Options** dialog box in the Windows Control Panel.
 3. In the **Languages** tab, set the required language for the menus and dialogs.
 4. In the **Advanced** tab and the **Regional Options** tab, set the required language.

Linux

- Linux hosts must have graphics capabilities to run Teamcenter installation tools.

For operating system requirements, see the Hardware and Software Certifications knowledge base article on Support Center.

- Linux hosts must have the **nslookup** utility available to ensure operation of the license server.
- Make sure Linux host names do not exceed 31 characters in length. Host names longer than 31 characters cause Teamcenter corporate server installation to fail during saving of the POM schema file in the *TC_DATA* directory.

Teamcenter installation tools do not require fully qualified domain names for host names. If your fully qualified domain name exceeds 31 characters, use the server short host name instead.

For more information, see the solutions document 002-7004480 on Support Center.

- Teamcenter Environment Manager (TEM) and Web Application Manager require the ISO8859-1 character set. Make sure this character set is available on your host.

Database

Teamcenter requires a relational database management system (RDBMS) for storing Teamcenter data. Before you install Teamcenter, you must install an Oracle database server or a Microsoft SQL Server database server.

If your database server is not a supported version, upgrade your database server to a supported version before you install Teamcenter.

Choose a database management system that suits the platforms of your Teamcenter servers and clients, and make sure your Teamcenter corporate server host has access to the database server.

If you use Oracle, set system parameters to recommended values to ensure adequate database performance.

Java Runtime Environment

Teamcenter Environment Manager (TEM) requires a supported 64-bit Java Runtime Environment (JRE). If a certified JRE is not available on the host, TEM cancels installation.

Before you launch TEM to install Teamcenter:

1. Download and install a certified 64-bit JRE.

For certified JRE versions, see the Hardware and Software Certifications knowledge base article on Support Center.

2. Set the **JRE_HOME** environment variable to the location of the supported JRE. After installation is complete, TEM no longer requires this variable.

Alternatively, you can launch TEM in a command prompt and specify the JRE location using the **-jre** argument:

```
tem -jre JRE-path
```

For example:

```
tem -jre c:\apps\jre1.8
```

Web tier support

Install the required software for the Teamcenter web tier you use:

- **Java EE web tier**

Java Runtime Environment (JRE)

Install a supported JRE on the host where you build Teamcenter web applications.

Java EE application server

Install a supported application server on the host where you deploy Teamcenter web applications.

- **.NET web tier**

Microsoft Internet Information Server (IIS)

Install IIS on your Teamcenter corporate server host and add the required role services.

Microsoft .NET framework

Install the .NET framework on all Teamcenter hosts.

If you use the Teamcenter Java EE web tier, install the following software:

Java Runtime Environment (JRE)

Install a supported JRE on the host where you build Teamcenter web applications.

Java EE application server

Install a supported Java EE application server on the host where you deploy Teamcenter web applications.

Some web application servers require special configuration for use with Teamcenter.

Web browser

A web browser is required if you use the following:

- Teamcenter online help
- Active Workspace
- Deployment Center

For these products, Teamcenter supports the following web browsers:

- Windows systems: Microsoft Edge, Mozilla Firefox, and Google Chrome
- Linux systems: Mozilla Firefox and Google Chrome

For supported browser versions, see the Hardware and Software Certifications knowledge base article on Support Center.

3. Upgrading templates

Templates

A *template* is a container that holds data model definitions. A template can contain any number of business objects, classes, lists of values, and business rules. Any Teamcenter feature that adds definitions to the data model has an associated template. In addition, templates can be supplied to you from another Teamcenter site, partner, or third party.

Caution:

You *must* have access to all templates used at your site before beginning the upgrade of the corporate server. This is very important. During the corporate server upgrade, you load all templates used by your site. Templates can only be loaded during the upgrade of the corporate server. They cannot be loaded after the upgrade. Failing to load all templates your site depends on causes problems in managing the data model and can result in loss of data, time, and money.

How templates are supplied

Templates are supplied by:

- The Teamcenter software kit
- Software kits for asynchronous Teamcenter releases and integrations
- Other Teamcenter sites, partners, or third parties

Templates supplied on the Teamcenter software kit

Teamcenter supplies templates to organize sets of definitions for specific industries, applications, or functional areas. Examples of templates provided by Teamcenter are Teamcenter Foundation and Wire Harness Configuration.

These templates are available in the software kit (the installation DVD or downloaded software). For example, the Teamcenter Foundation feature and the Wire Harness Configuration feature are both installed using the software kit, both add definitions to the data model, and both have associated templates found in the software kit.

Templates supplied by asynchronous Teamcenter releases

Some Teamcenter features are released asynchronously and are distributed separately from the Teamcenter software kit (the installation DVD or downloaded software) and are found on their own software kit. An example is the CATIA integration.

Templates supplied by other Teamcenter sites, partners, or third parties

You can use templates created by other sites, partners, or third parties.

Loading templates

Before you upgrade the corporate server, you must have access to all templates on which your site depends.

- Templates supplied by Teamcenter on the installation kit are automatically detected by Teamcenter Environment Manager (TEM) and loaded during the upgrade.
- Templates supplied asynchronously by Teamcenter must be detected and loaded during the upgrade of the corporate server. This is done using the **Upgrade Database Features** panel in TEM during the upgrade.

You must have access to these templates before beginning the corporate server upgrade. Therefore, you cannot begin your upgrade until all asynchronous features that you use are released.

- Templates supplied by another Teamcenter site, partner, or third party must be loaded manually during the upgrade of the corporate server. This is done using the **Upgrade Database Features** panel in TEM during the upgrade.

You must have access to these templates before beginning the corporate server upgrade.

Upgrade Database Features panel

- Templates are loaded into your environment during your corporate server upgrade using the **Upgrade Database Features** panel in TEM.

All features available from the software kit are listed. Those that you have not installed are grayed-out.

- Templates supplied by Teamcenter asynchronous from the kit, templates supplied by another site, partner, or third party are not listed in the **Upgrade Database Features** panel until you manually load them. Click the **Browse** button and select the templates.
- After you select the template, it is listed in the **Upgrade Database Feature** panel.

Business Modeler IDE

The Business Modeler Integrated Development Environment (IDE) is a tool for customizing the data model of your Teamcenter installation. Use the IDE to create business objects, classes, attributes, lists of values (LOVs), and rules.

The Business Modeler IDE is built on top of the Eclipse platform. Eclipse is a generic platform for tool development that is extended via its plug-in and extension point technology.

You can install the Business Modeler IDE two ways:

- As a stand-alone application

This method installs the Eclipse platform.

- As part of an existing Eclipse environment

4. Using the Upgrade Assistant

Teamcenter provides a tool, the *Upgrade Assistant* utility, to help you determine impact to your server side customization extension applications when you upgrade to a target release. It evaluates whether custom code needs to be reworked before upgrade. You can evaluate the tool's recommended remediation, for example, ensuring API stability or removing calls to deprecated or obsolete APIs.

This tool currently reports only the usage of deprecated and obsolete ITKs in server side customization applications. The utility takes server side customization application libraries (**dlls**) as input and runs a **dumpbin** report on the library looking for import symbols. The import symbols are then analyzed with the deprecated and obsoleted API in the target release to produce a **csv** file report indicating whether rework is needed immediately (if the application is using an obsolete ITK) or future rework should be planned (if the application is using a deprecated ITK).

The *What's changed in Teamcenter APIs* reference on Support Center identifies replacements for deprecated and obsolete APIs. This reference also provides replacement information for other deprecated and obsolete artifacts like functions and macros. Use this reference when updating your customizations in preparation for upgrade.

Run the Upgrade Assistant as part of your upgrade planning, to help identify and correct customization problems before you begin upgrading your Teamcenter environment. You can run this utility before you download your target upgrade version of Teamcenter.

Download the Upgrade Assistant

1. Log on to Support Center and browse to the Teamcenter 14.3 downloads page.
2. Browse to **Additional Downloads**→ **Tools for Teamcenter**→ **Tools for Teamcenter Upgrade Assistant**.
3. In the resulting list of downloads, find the Upgrade Assistant, and then download the **Tc14.3_UpgradeAssistantITKReporter.zip** package for your platform (Windows or Linux).

Install the Upgrade Assistant (Windows)

1. Make sure a supported version of Microsoft Visual Studio is installed on your local machine. For certified versions of Visual Studio, see the Hardware and Software Certifications knowledge base article on Support Center.
2. Make sure the following system environment variables are set:

MSDEV_HOME

Set to the Visual Studio installation directory, for example, **set MSDEV_HOME=C:\apps\MVS16\VC**.

PATH

Make sure this value includes your Teamcenter Perl install directory, for example, **set PATH=C:\apps\tc\tc14\TR\perl\bin;%PATH%**.

- Expand the **Tc14.3_UpgradeAssistantITKReporter.zip** package to a local directory.
- Open a Teamcenter command prompt by choosing **Start→All programs→Teamcenter 14→Tc-config-name Command Prompt**.¹
- Change to the directory in which you expanded the Upgrade Assistant package:

```
cd upgrade-assistant-dir
```

- Type the following command to launch the Upgrade Assistant and view its arguments and options:

```
Tc14.3_UpgradeAssistantITKReporter\ITKReporter\bin\TcUpgradeAssistantITKReporter.bat -h
```

Install the Upgrade Assistant (Linux)

- Expand the **Tc14.3_UpgradeAssistantITKReporter.zip** package to a local directory.
- Change to the directory in which you expanded the Upgrade Assistant package:

```
cd upgrade-assistant-dir
```

- Run the following command to launch the Upgrade Assistant and view its arguments and options:

```
./ITKReporter/bin/TcUpgradeAssistantITKReporter.sh -h
```

Generate an Upgrade Assistant report

The Upgrade Assistant utility accepts the following arguments:

Argument	Description
apps	Specifies a single path or a sequence of comma-separated file paths, or a text file containing a list of custom applications (absolute path of one application per line) or a directory path containing a list of custom applications.
from_release	Specifies your current Teamcenter version, for example, 14.0 . This argument is optional. If this argument is not provided, the utility reads the Teamcenter version from the environment variables in the Teamcenter command prompt.
out	Specifies the output file (csv format) and path for the Upgrade Assistant report. For example:

¹ Alternatively, you can set Teamcenter environment variables in a standard command prompt by typing **TC_DATA\tc_profilevars**, replacing **TC_DATA** with the path to your Teamcenter data directory.

Argument	Description
	D:\Temp\mycustom_13_14.csv
h	Displays help for the utility.

To generate an Upgrade Assistant report, perform the following steps:

1. Open a Teamcenter command prompt.

On Windows systems, click **Start→All programs→Teamcenter 14→Tc-config-name Command Prompt**.

2. Change to the *upgrade-assistant-dir\Tc14.3_UpgradeAssistant\ITKReporter\bin* directory (on Windows systems) or *upgrade-assistant-dir/Tc14.3_UpgradeAssistant\ITKReporter/ITKReporter/bin* (on Linux systems).

3. Type the following command:

- **Windows:**

```
TcUpgradeAssistantITKReporter.bat -apps=custom-path -from_release=current-ver
-out=report-path
```

- **Linux:**

```
TcUpgradeAssistantITKReporter.sh -apps=custom-path -from_release=current-ver
-out=report-path
```

In the appropriate command, replace *custom-path* with the path or paths to your custom application **dll** files. Replace *current-ver* with your current Teamcenter version. (This argument is optional.²) Replace *report-path* with the location in which to generate the report, in **csv** format.

Examples (Windows):

- To generate an Upgrade Assistant report in the current directory:

```
TcUpgradeAssistantITKReporter.bat -apps=D:\\myDir\\MyCustom
```

- To generate an Upgrade Assistant report into a specific directory:

```
TcUpgradeAssistantITKReporter.bat -apps=D:\\myDir\\MyCustom -out=D:\\
\\Temp\\mycustom_13_14.csv
```

² If **from_release** is not specified, the Teamcenter version is read from the environment variables in the Teamcenter command prompt.

- To generate an Upgrade Assistant report for a Teamcenter version other than the version specified in the Teamcenter command prompt:

```
TcUpgradeAssistantITKReporter.bat -apps=D:\\myDir\\MyCustom  
-from_release=13.0 -out=D:\\Temp\\mycustom_13_14.csv
```

Examples (Linux):

- To generate an Upgrade Assistant report in the current directory:

```
TcUpgradeAssistantITKReporter.sh -apps=/home/myusername/myDir/  
MyCustom.so -from_release=11.0 -out=<output directory>\  
\\mycustom_13_14.csv
```

- To generate an Upgrade Assistant report into a specific directory:

```
TcUpgradeAssistantITKReporter.sh -apps=/home/myusername/myDir  
-from_release=11.0 -out=<output directory>\\mycustom_13_14.csv
```

- To generate an Upgrade Assistant report for a Teamcenter version other than the version specified in the Teamcenter command prompt:

```
TcUpgradeAssistantITKReporter.sh -apps=/home/myusername/myfile.txt  
-from_release=11.0 -out=<output directory>\\mycustom_13_14.csv
```

Part II: Build the Teamcenter upgrade test environment



Begin preparing for your test environment by verifying your target machines meet the system requirements for Teamcenter 14.3.

Install the latest version of the Siemens Digital Industries Software License Server and install your updated license file. Also, upgrade your database server to a version supported for Teamcenter 14.3.

5. Preparing the Teamcenter network

Install the Siemens License Server

Before you upgrade Teamcenter, you must download and install the **Siemens License Server** to distribute licenses to Teamcenter hosts. Even if you already use the Siemens License Server with your current version of Teamcenter, you must install the version of the License Server supported by the version of Teamcenter to which you upgrade.

For the version of the Siemens License Server certified with Teamcenter 14.3, see the Hardware and Software Certifications knowledge base article on Support Center.

Download and install the Siemens License Server:

1. Open Support Center:

<https://support.sw.siemens.com>

2. Under **Product Centers**, find **Siemens License Server**.

Caution:

Make sure you download **Siemens License Server**, *not* **Siemens PLM Licensing**.¹

3. In the Siemens License Server product center, click **Downloads**, and then download the certified version of the Siemens License Server.
4. Install the License Server according to the *Siemens Digital Industries Software License Server Installation Instructions* available from the Siemens License Server downloads page.
5. On your designated Teamcenter corporate server host, set the following system environment variables:

SPLM_LICENSE_SERVER

Set to the location of the Siemens License Server:

port@host

Replace *port* with the port number and *host* with the machine name of the License Server, for example, **29000@tchost**.

¹ Siemens PLM Licensing is no longer supported by Teamcenter. The Siemens License Server is the currently supported license server.

The value of this variable is designated as the default local license server during corporate server upgrade.

If you upgrade Teamcenter using TEM, TEM verifies that the specified license server exists and is running a supported version of the Siemens License Server. If the configured license server is not valid, the upgrade is stopped until a valid license server is installed.

TCP_NODELAY

Set to a value of **1** on the License Server host. This helps optimize logon time when launching Teamcenter.

6. Install Teamcenter licenses on the License Server according to the information provided to you by Siemens Digital Industries Software support.

The [Siemens License Server downloads page](#) contains additional links to documentation, Knowledge Base articles, and videos about installing and maintaining the License Server.

Caution:

The License Server must be running and two or more seats must be available on that license server during Teamcenter server installation. Otherwise, database creation fails because the **make_user** utility cannot create the required users in the database.

Upgrading database servers

Supported database server versions

If your database server version is not a version certified for Teamcenter 14.3, you must upgrade your database server before you upgrade Teamcenter.

Siemens Digital Industries Software supports Teamcenter 14.3 with Oracle and Microsoft SQL Server databases. For supported versions of Oracle and Microsoft SQL Server, see the Hardware and Software Certifications knowledge base article on Support Center.

Configuring character settings in non-English locales

To ensure correct display and processing of Teamcenter data, you must set required values in your system environment and your Teamcenter configuration.

If your Teamcenter servers run Linux and use the Unicode UTF-8 character set, you must configure your operating system and your Teamcenter configuration to use Unicode UTF-8, observing the following considerations during Teamcenter upgrade:

- An existing Teamcenter installation must not configure Unicode UTF-8 character set support with an upgrade to Teamcenter 14.3. An existing Teamcenter installation is required to complete an upgrade to Teamcenter 14.3 under its existing database character set encoding.

- After upgrade is completed, you must contact your database vendor for processes and tools to convert your existing database character set encoded data to Unicode UTF-8 character set encoded data for storage to a new database.

For more information about environment settings for your locale, platform, and character set, see UTF-8 settings for Windows or Linux systems, or non-UTF-8 settings for Windows or Linux systems in the Teamcenter installation guides.

Set shell limits and parameters for Oracle on Linux

Overview of shell limits and parameters

Oracle RDBMS uses extensive Linux resources such as shared memory, swap memory, and semaphore for interprocess communication. Inadequate parameter settings cause problems during installation and startup. Increasing the volume of data stored in memory reduces disk I/O activity and improves database performance.

The Oracle RDBMS installation program displays warnings if kernel parameters are not adequate. To avoid warnings and errors during or after installation, make sure kernel parameters meet the recommended settings for typical environments described in the following topics.

Before you install Oracle RDBMS, set initial parameters as described in Oracle documentation, and then adjust parameters according to available system memory. Set the **ulimit** parameter to **unlimited**.² Then, set the **kernel parameters** to recommended Teamcenter values for your operating system.

If you previously tuned kernel parameters for other installed applications to levels that meet or exceed the values recommended for Teamcenter, keep those existing values.

The parameter settings recommended herein are *minimum* values. For production database systems, Oracle recommends you tune values to optimize system performance. For information about performance tuning, see:

- Documentation for your operating system
- Teamcenter installation documentation on Support Center

Set SUSE Linux shell limits

1. Increase shell limits for the **oracle** user to the minimum values listed in the following table by adding the following lines to the **/etc/security/limits.conf** file:

oracle	soft	nproc	2047
oracle	hard	nproc	16384
oracle	soft	nofile	1024
oracle	hard	nofile	65536

² The **ulimit** parameter specifies a maximum number of processes per user.

Do not change the shell limit values if they were set for another program and the values are greater than the levels Oracle requires.

SUSE Linux shell limit	Item in limits.conf	Minimum hard limit
Maximum number of open file descriptors	nofile	65536
Maximum number of processes available to a single user	nproc	16384

2. Add or edit the following lines in the **/etc/pam.d/login** file:

```
session required /lib64/security/pam_limits.so
session required pam_limits.so
```

3. Change the **oracle** user default shell startup file:

- For the Bourne, Bash, or Korn shell, add the following lines to the **/etc/profile.local** file:

```
if [ $USER = "oracle" ]; then
    if [ $SHELL = "/bin/ksh" ]; then
        ulimit -u 16384
        ulimit -n 65536
    else
        ulimit -u 16384 -n 65536
    fi
fi
```

- For the C shell (csh or tcsh), add the following lines to the **/etc/csh.login.local** file:

```
if ( $USER == "oracle" ) then
    limit maxproc 16384
    limit descriptors 65536
endif
```

Upgrade an Oracle server and database

Export an Oracle database

Windows systems:

1. Log on to the Oracle server as an administrator user.
2. Export the contents of your Teamcenter Oracle database to the dump file:

```
ORACLE_HOME\bin\expdp db-user/password full=y dumpfile=file-name.dmp
logfile=export.log
```

Replace *db-user* with the Teamcenter database user account name; replace *password* with the database user account password; replace *file-name* with the full path and name of the dump file to contain the exported data; replace *export* with the name of the log file to contain export output.

3. Store the dump file in a safe place.

Linux systems:

1. Either log on to the Oracle server as **oracle** or switch the user to **oracle**:

```
su - oracle
```

2. Set the **PATH** environment variable to include the Oracle **bin** directory:

```
export PATH=$PATH:ORACLE_HOME/bin
```

3. Manually set the shared library path for Linux:

```
export LD_LIBRARY_PATH=${LD_LIBRARY_PATH}:${ORACLE_HOME}/lib
```

4. Export the contents of the Teamcenter Oracle database to the dump file:

```
ORACLE_HOME/bin/exp db-user/password full=y file=file-name.dmp
log=export.log
```

Replace *db-user* with the Teamcenter database user account name; replace *password* with the database user account password; replace *file-name* with the name of the dump file to contain the exported data; replace *export* with the name of the log file to contain export output.

5. Store the dump file in a safe place.

If you have multiple databases, repeat this procedure for each database.

Caution:

Siemens Digital Industries Software strongly recommends backing up the dump file on tape or another disk. If the dump file becomes corrupted or lost, all data from the existing database is lost.

Terminate Oracle sessions on Windows systems

Stop the listener process

1. Log on to the operating system as a user with administrator privileges.

2. Open the **Services** dialog box in the Windows Control Panel.
3. Select the Oracle TNS listener services (**Oracle~~release-ID~~TNSListener**) and click **Stop**.

Shut down an Oracle database

Shut down Oracle using Windows Control Panel

1. Log on to the operating system as a user with administrator privileges.
2. Open the **Services** dialog box in the Windows Control Panel.

Windows displays the Services window.

3. Select the **OracleServiceSID** service.

Replace *SID* with the system identifier of the database instance.

4. Click **Stop**.

Shut down Oracle using SQL*Plus

1. Log on to the operating system as a user with administrator privileges.
2. Start the Oracle SQL*Plus utility:

```
sqlplus sys/password@Oracle-SID as sysdba
```

Replace *password* with the password for the **sys** user account.

Oracle starts the Oracle SQL*Plus utility.

The **sys** user must be in the Oracle **sysdba** group for the Oracle system identifier (SID) used by Teamcenter. To connect as internal (without a password), the account must be part of the **ORA_DBA** local group in Windows.

3. Shut down the database instance by typing the following command:

```
shutdown
```

4. Exit SQL*Plus:

```
exit
```

Terminate Oracle sessions on Linux systems

Before installing a new version of Oracle, you must terminate all Oracle sessions and Oracle processes.

1. Either log on to the Oracle server as **oracle** or switch the user to **oracle** as follows:

```
su - oracle
```

2. Set the **ORACLE_HOME** environment variable to point to the location of the Oracle files. For example:

```
export ORACLE_HOME=/u01/app/oracle/product/oracle-version
```

Replace the path with the system path to the Oracle files.

3. Define **ORACLE_HOME/bin** in the **PATH** variable:

```
export PATH=${PATH}:${ORACLE_HOME}/bin
```

4. Manually set the shared library path on Linux:

```
export LD_LIBRARY_PATH=${LD_LIBRARY_PATH}:${ORACLE_HOME}/lib
```

5. If a **tnslsnr** listener process is running, terminate it. For example:

```
${ORACLE_HOME}/bin/lsnrctl stop listener-name
```

Replace *listener-name* with the name of the listener process.

6. Shut down all Oracle database instances using the **dbshut** utility. Shut down database instances listed in the **oratab** file:

```
${ORACLE_HOME}/bin/dbshut
```

Back up an Oracle installation

If you are upgrading to the certified Oracle version, back up the existing Oracle installation.

Backing up your Oracle installation before upgrading is strongly recommended. Failure to back up existing data could result in loss of data if problems occur during the upgrade process.

Back up the following files and directories:

- The Oracle home directory on each installed workstation.
- The directories containing database files for each configured database.

- The Oracle Net **listener.ora** and **tnsnames.ora** configuration files in the **/etc** directory.

These are the only Teamcenter directories affected by Oracle installation. If you created other directories containing data used by Oracle, such as an administration script directory, you should also back up these directories.

Upgrading an Oracle server

Upgrade the Oracle server

Upgrade your Oracle server by one of the following methods:

- *Upgrade using the Oracle installer*
- *Upgrade by uninstalling/reinstalling Oracle*

Upgrade using the Oracle installer

1. Launch the Oracle installer to install a certified version of Oracle server.
2. When the Oracle installer prompts you to upgrade existing databases, enter the required information about the databases you want to upgrade.

Installing an Oracle server is described in the Teamcenter installation guides for Windows and Linux.

Upgrade by uninstalling/reinstalling Oracle

1. Remove existing Oracle databases.
2. Uninstall all existing Oracle server software.
3. Install a certified version of Oracle server.

Installation of an Oracle server is described in the Teamcenter installation guides for Windows or Linux.

4. After Oracle installation is complete, import your Teamcenter database from the Oracle dump file into the new Oracle database. Enter the following command on a single line:

```
ORACLE_HOME\bin\imp db-user/password fromuser=db-user touser=db-user
file=file-name.dmp log=import.log
```

Replace *db-user* with the Teamcenter database user account name, *password* with the database user account password, *file-name* with the full path and name of the dump file that contains the exported data, and *import* with the name of the log file.

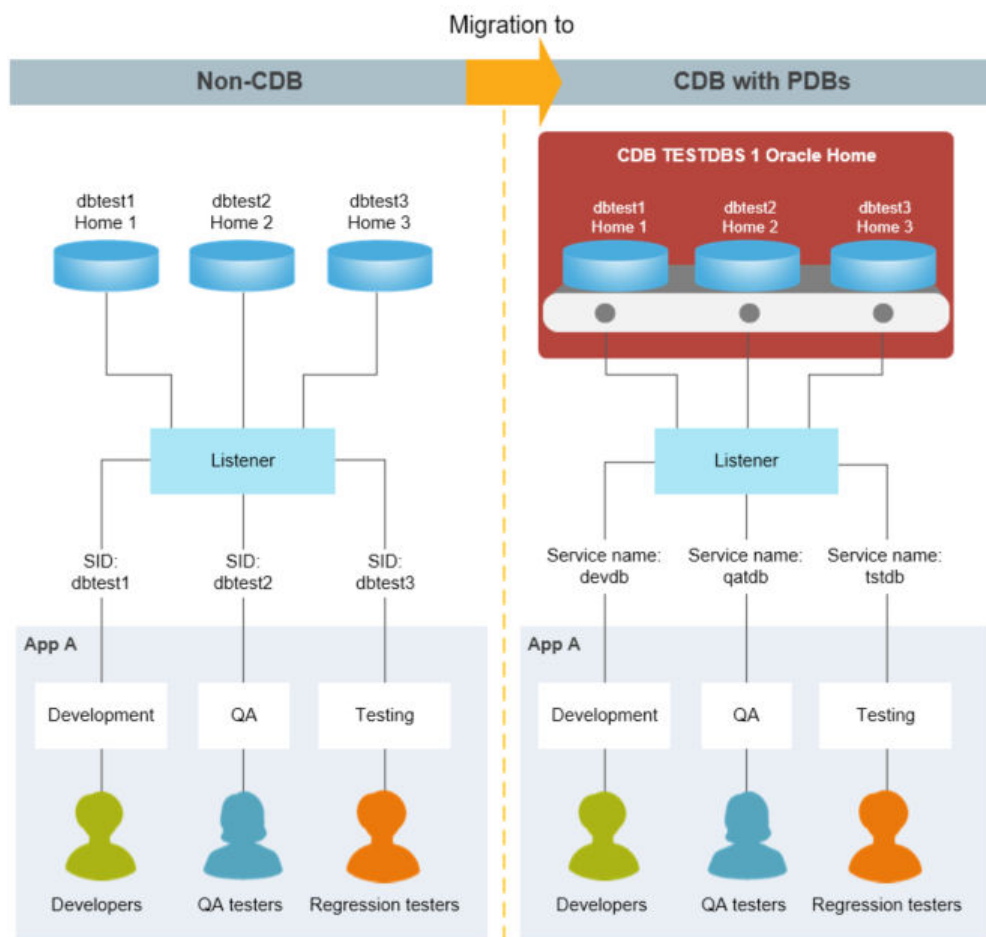
Migrate a non-CDB database to a CDB database

Teamcenter supports Oracle's **multitenant database architecture** if you use Oracle 12c or later. A multitenant architecture is deployed as a Container Database (CDB) with one or more Pluggable Databases (PDB).

A *Container Database* (CDB) is similar to a conventional (non-CDB) Oracle database, with familiar concepts like control files, data files, undo, temp files, redo logs, and so on. It also houses the data dictionary for objects owned by the root container and those that are visible to databases in the container.

A *Pluggable Database* (PDB) contains information specific to the database itself, relying on the container database for its control files, redo logs and so on. The PDB contains data files and temp files for its own objects, plus its own data dictionary that contains information about objects specific to the PDB. From Oracle 12.2 onward a PDB can and should have a local undo tablespace.

You can **migrate a non-CDB database to a CDB database** using Oracle tools. The following example illustrates the database architectures before and after migration.



Teamcenter supports CDB and non-CDB databases. Be aware that **Oracle has deprecated support for non-CDB databases** and may discontinue support after Oracle 19c.

If you migrate a non-CDB Teamcenter database to a CDB database, you must perform the migration *after* you upgrade to Teamcenter 14.3.

Upgrade Microsoft SQL Server databases

Upgrading a Microsoft SQL Server database server with an existing Teamcenter database requires exporting the contents of the existing database, installing the new Microsoft SQL Server database server, and importing the database into the new server.

Microsoft describes several methods for transferring databases between servers in the article titled *How to move databases between computers that are running SQL Server* at <https://support.microsoft.com>.

Siemens Digital Industries Software recommends a backup and restore method for transferring an existing Teamcenter database to a new Microsoft SQL Server server. One advantage of this method is the simplicity of a single backup file, rather than multiple files (**.mdf**, **.ndf**, and so on), as with an attach/detach method. A single backup file also minimizes potential compatibility mode issues between versions of Microsoft SQL Server.

Using Data Transformation Services (DTS) for transferring a database is the least recommended method because it is relatively more time-consuming and requires communication between the old and new database servers. In contrast, other methods for database transfer require only a means to move export files from one server to the other.

Siemens Digital Industries Software recommends creating the Teamcenter database user in the new target database *before* importing the existing database into the new server.

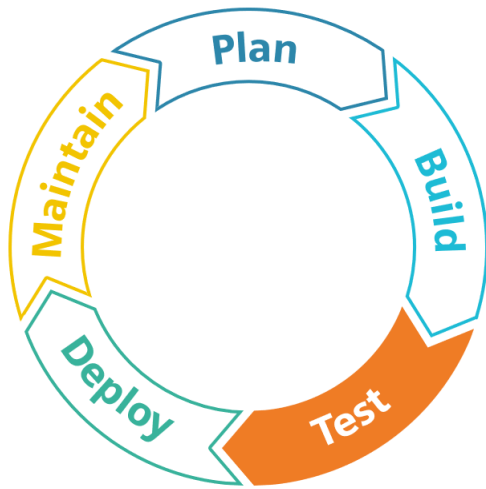
For transferring logins and passwords to a new Microsoft SQL Server instance, Microsoft provides an article titled *How to transfer logins and passwords between instances of SQL Server* at <https://support.microsoft.com>.

To ensure the owner of the database is the new server's Teamcenter database user account rather than the old server's Teamcenter database user account,³ enter the following commands:

```
exec sp_changedbowner 'sa'
exec sp_changedbowner 'Tc-db-user'
```

³ Internal integer IDs used by SQL Server can differ between sites.

Part III: Test the Teamcenter upgrade



Before you upgrade your production systems to the target Teamcenter version, perform an upgrade on a test system. The upgrade process consists of the following tasks:

1. **Complete applicable pre-upgrade tasks.**
2. **Upgrade your custom Business Modeler IDE templates.**
Upgrade a template project to the current data model format.
3. **Upgrade a test system.**
 - a. **Create an upgrade testing environment.**
 - b. **Upgrade the test environment.**
4. **Complete applicable post-upgrade tasks.**
5. **Update microservice framework and Active Workspace or add microservice framework and Active Workspace** to an environment that does not yet have them.

Deploy the Teamcenter upgrade after you have completed these steps.

You must upgrade your template project in the Business Modeler IDE before you upgrade a test database or your production systems because the XML formats of template files can change between product releases. Your custom template project must be modified to fit the new XML format.

Always test the upgrade process on a test system that reflects your production system before you begin upgrading your production system. Upgrading a test system can reveal issues that may prevent an upgrade from completing successfully. After you resolve these issues on the test system, you can proceed with upgrading your production system. This helps minimize downtime for your production environment.

Note:

When you upgrade to the latest version of Teamcenter, existing Change Viewer Classic change objects are not upgraded. Additional steps are required to upgrade your Change Viewer Classic change objects to Change Manager.

If your Teamcenter installation includes Teamcenter Integration for I-deas, make sure you run the **TcII.bat** file (on Windows systems) or the **TcII.sh** script (on Linux systems) before you launch Teamcenter Environment Manager (TEM) to upgrade the corporate server. This sets the **CLASSPATH** variable and other variables required to enable TEM to find installation files for Teamcenter Integration for I-deas.

The **TcII.bat/sh** script is on the Teamcenter Integration for I-deas software kit.

6. Test upgrading a Teamcenter corporate server

Pre-upgrade tasks

Install C Shell on Linux servers

Before upgrading a corporate server that is hosted on a Linux machine, you must verify that C Shell is installed.

If it is not installed, the Pre-Upgrade Diagnostic test fails at the Volume Access Check step with an error similar to the following:

```
Name: Volume Access Check
Description:
Test whether the FSC is up and running
Analysis:
An attempt to contact the FSC running at http://SERVER:4544 failed.
Check that the FSC is running and configured correctly.
```

If you see this error, install C Shell and rerun the Pre-Upgrade Diagnostic test.

Verify required groups and roles

Before you begin upgrade, verify the following groups exist in your database:

```
dba
system
Project Administration
Validation Administration
```

Also, verify the following roles exist in the specified groups.

Role	Group
Project Administrator	Project Administration
Validation Administrator	Validation Administration

Upgrade may fail if these groups and roles are not present.

Verify required character set

You must have the same locale installed on your Teamcenter host as you use to communicate with your database server, and the database server must support this locale as well.

On Linux systems, Teamcenter installation tools, verify that the required character set is loaded by running the **locale -a** command in a shell. If the output does not list the required character set, you must add this character set before you upgrade Teamcenter.

1. Set or export the **LC_ALL** environment variable by typing **LC_ALL=character-set** or the equivalent command for your platform.
2. Verify the setting using the **echo** command or equivalent. Make sure the correct value for **LC_ALL** is displayed.
3. Run the **locale** command and make sure the **LANG** variable and all the **LC_x** variables are set the same as **LC_ALL**.
4. If **LANG** is still set to **C**, manually export **LANG** to be the same value as **LC_ALL**.
5. Launch Teamcenter Environment Manager (**tem.sh**) from the current shell.

Alternatively, your system administrator may modify the date file (named **TIMEZONE** in the **etc** directory), which can preset this environment, so every time you log on and launch a shell, the environment is preset.

The recommended method, however, is to log on to the system using the Common Desktop Environment (CDE) with the minimum required locale by choosing **Option→Language→character-set** during logon.

If the required character set is not loaded on your machine, contact your system administrator to have it installed before you install the GM Overlay.

This requirement is necessary because current Teamcenter versions use XML files rather than **.dat** files and associated scripts. Because of this, GM Overlay data is transformed from **.dat** files into XML files.

To read and parse the XML files correctly, the system must be able to process non-English (non-ASCII) locale characters. To facilitate this, the system must be first loaded with the fonts for that locale.

Verify site consistency

If your production database contains any transfer locks or site inconsistency objects, the test upgrade will not start. When TEM performs pre-upgrade diagnostics, the site consistency check fails and TEM does not allow the upgrade to proceed until site inconsistencies are corrected. TEM provides a tool to fix site inconsistencies, but the tool runs only on Multi-Site environments.

Before you copy a production database for a test upgrade, Siemens Digital Industries Software strongly recommends performing the following steps:

1. Perform upgrade diagnostics on your production database. Begin an upgrade TEM, proceed to the **Pre-upgrade Diagnostics** panel, and then click **Run** to perform pre-upgrade diagnostics.
2. When pre-upgrade diagnostics complete, click **Review** to view the pre-upgrade diagnostics report.
3. Perform any steps recommended in the report to resolve site inconsistency errors.

Terminate Teamcenter sessions

Prior to upgrade, you must terminate Teamcenter sessions if:

- You are reinstalling or upgrading Teamcenter executables by overwriting an existing Teamcenter data directory. The Teamcenter installation procedure cannot overwrite files when they are in use.
 - You are upgrading a Teamcenter database.
 - You are migrating an Oracle database to a Windows database server.
1. Instruct all users to check in all Teamcenter business objects, and then close and log off of Teamcenter sessions, including **tcserver** processes.
 2. Open a Teamcenter command prompt:

Windows systems:

From the **Start** menu, choose **Programs**→**Teamcenter**, and open a command prompt.

Linux systems:

Enter the following commands:

```
TC_ROOT=/usr/Siemens/Teamcenterversion; export TC_ROOT
TC_DATA=/usr/Siemens/Teamcenterversion/teamcenterdata; export
TC_DATA
. $TC_DATA/tc_profilevars
```

Replace *version* with the Teamcenter version.

This example assumes that Teamcenter is installed under the **usr/Siemens/Teamcenterversion** directory.

Sourcing the **tc_cshvars** file creates a **csh** subshell in which Teamcenter environment variables are set.

On operating systems that do not support sourcing, enter the following commands to set Teamcenter environment variables:

```
#!/bin/bash
export TC_ROOT=/usr/Siemens/Teamcenterversion/tc_root
export TC_DATA=/usr/Siemens/Teamcenterversion/tc_data
bash --init-file $TC_DATA/tc_profilevars
```

Replace *version* with the Teamcenter version.

3. Use the **clearlocks** utility to check for nodes connected to the database and remove locks on the database:

Windows systems:

```
%TC_BIN%\clearlocks -u=Tc-Oracle-user -p=Tc-Oracle-user-password -g=dba
-assert_all_dead
```

Linux systems:

```
$TC_ROOT/bin/clearlocks -node_names
```

4. On Linux systems, note the node names returned, and then type the following command for each node name returned:

```
$TC_ROOT/bin/clearlocks -assert_dead node-name
```

Replace *node-name* with a returned node name.

5. Stop all Teamcenter services, including FMS.

Back up existing Teamcenter data

If you upgrade a Teamcenter database, back up existing Teamcenter data.

Caution:

Back up the database, Teamcenter data directory, and all Teamcenter volume directories to an external backup device before performing an upgrade. This provides a safeguard against data loss in case problems occur during the upgrade.

Back up the following directories:

- The Teamcenter application root directory on each installed workstation
- The Teamcenter data directory for each configured database

- The Teamcenter volume directories for each configured database

These are the only directories affected by Teamcenter installation. If you created other directories that contain data used by your existing Teamcenter installation, such as a separate POM transmit schema directory, Siemens Digital Industries Software recommends that you back up these directories as a precautionary measure.

Clean unused columns from the database

If you use an Oracle database, upgrade performance may decrease when dropping columns from a large Teamcenter class. This can affect overall upgrade time.

To minimize overall upgrade time, you can mark a column to be dropped as unused by typing the following SQL statement:

```
ALTER TABLE table-name SET UNUSED column-name
```

This statement marks the column as unused and hides it from any SQL used on that table. Commands like **DESCRIBE** *table_name* or **SELECT * from** *table-name* will not show the column. Any column that is marked as unused is not displayed in queries or data dictionary views and its name is removed so that a new column can reuse the same name. All constraints, indexes, and statistics defined on the column are also removed.

To physically clean up unused columns and reclaim space, use the **install** utility with the **clean_unused_columns** argument:

```
install -clean_unused_columns Tc-Oracle-user password dba
```

Depending on the number of unused columns in the database and size of their related tables, this command may generate large **redo** logs, especially if large tables are involved. Siemens Digital Industries Software recommends you adjust the size of the **redo** logs appropriately before attempting to use this utility, and make sure you have exclusive access to the schema, for example, during the maintenance window.

Estimate tablespace requirements for upgrade

Upgrading to Teamcenter 14.3 temporarily requires additional database tablespace.

To improve system performance, Teamcenter converts **POM_Stub** and **ImanExportRecord** objects to lightweight objects (LWOs). This changes the persisted schema, and these schema changes require some of the persisted data (including the **POM_Object** class table) to be reorganized. For large databases, these changes could take many days to complete by the prior approach of updating each row of the table. The Teamcenter upgrade process makes a copy of the table, inserting and modifying rows into the copy during the copying process. This upgrade process then replaces the original data tables with the modified data tables. This upgrade performance improvement requires more disk space, but only for the duration of the upgrade process.

Make sure your system has sufficient disk space to accommodate these needs during the upgrade process.

Oracle databases:

1. **Back up existing Teamcenter data.**
2. Determine the tables affected by the upgrade to Teamcenter 14.3.
 - a. Enter the following SQL command:

```
SELECT pname FROM PPOM_CLASS
WHERE BITAND(pproperties, 8192) <> 0
UNION
SELECT pname FROM PPOM_CLASS
WHERE BITAND(pproperties, 32768) <> 0;
```

- b. To the resulting list of tables, add the **PIMANEXPORTRECORD** and **PPOM_STUB** tables, which are also affected by the upgrade process.
3. Determine the amount of space consumed by each affected table.

For each table identified in step 2 (including **PIMANEXPORTRECORD** and **PPOM_STUB**), enter the following SQL command:

```
WITH cte1
  AS (SELECT table_name
      FROM user_tables
      WHERE table_name = 'TABLE-NAME'),
  cte2
  AS (SELECT SUM(blocks) * 8 / 1024 MB
      FROM dba_extents,
           cte1

WHERE owner = (SELECT SYS_CONTEXT('userenv', 'current_schema')
                FROM dual)
              AND segment_name = cte1.table_name)
SELECT ROUND(SUM(mb), 0) || ' MB' AS ESTIMATED_SPACE
FROM cte2;
```

Replace *TABLE-NAME* with the name of the given table.

4. Determine the amount of additional free space you need to complete the upgrade to Teamcenter 14.3.

Table	Space needed	Where space is needed
Tables found in step 2a	1.5 times current space	Current tablespace of the given table
PIMANEXPORTRECORD	1.5 times current space	Database user's default tablespace
PPOM_STUB	1.5 times current space	Database user's default tablespace

5. Optionally, specify locations for new tablespaces and indexes.

- If you want to ensure tables are rebuilt in a specific tablespace, set the **TC_UPGRADE_TABLE_STORAGE** environment variable to the appropriate tablespace name before you launch TEM to begin the upgrade.
- If you want to ensure indexes for the upgraded tables are rebuilt in a specific tablespace, set the **TC_UPGRADE_INDEX_STORAGE** environment variable to the appropriate tablespace name before you launch TEM to begin the upgrade.

This applies *only* to tables identified in step 2a. The **PIMANEXPORTRECORD** and **PPOM_STUB** tables can only be rebuilt in the default tablespace.

After the upgrade is complete, you can move tables back to their original tablespaces.

Microsoft SQL Server databases:

1. **Back up existing Teamcenter data.**
2. Change the recovery model of the SQL Server database to **SIMPLE**.

After the upgrade, set the recovery model back to its previous value.

For information about changing the recovery model, see Microsoft SQL Server help:

<https://msdn.microsoft.com>

Perform a test upgrade. If the test upgrade fails during database upgrade, increase the available disk space for tablespaces and attempt the test upgrade again.

Upgrade a template project to the current data model format

If you have installed a new version of the Business Modeler IDE, you can use a project from the previous version. But first you must upgrade the project to the new data model format. This upgrade is necessary because the XML format used for data model files can change between product releases, and the project must be adjusted to fit the new XML format.

You can upgrade a project three ways:

- **Welcome** window

When you first open the Business Modeler IDE after installing it, the **Welcome** window is displayed. Click the **Upgrade your BMIDE template from a previous Teamcenter release** link in this window to run the import wizard. This imports your template into the new version of the Business Modeler IDE.

- Import wizard

If your template project is not already in the workspace, import it into the new version of the Business Modeler IDE:

1. Choose **File→Import**.
2. In the **Import** dialog box, choose **Business Modeler IDE→Import a Business Modeler IDE Template Project**.

While importing the project, the Business Modeler IDE automatically upgrades the project to the new data model format.

- **Re-run Template Project Upgrade** wizard

If your template project is already in the workspace, upgrade it to the new version of the Business Modeler IDE:

1. On the menu bar, choose **BMIDE→Upgrade Tools→Re-run Template Project Upgrade Wizard**.

The wizard runs.

2. In the **Template Project Upgrade** dialog box, click the arrow in the **Project** box to select the project to upgrade.
3. Click **Finish**.

The project is upgraded to the new data model format. The **Console** view displays success or failure messages for the upgrade.

After upgrade, open the **Project Files** folder and check for any error or warning messages in the log in the **output\upgrade** folder.

Caution:

After a template project is upgraded, it cannot be used for installation or upgrade in a previous version of Teamcenter. To find the version the template has been upgraded to,

open the **dependency.xml** file in the **extensions** folder of the template project and view the **currentTemplateVersion** value.

Upgrading the custom template may be part of a larger process when you upgrade to the latest version of Teamcenter:

1. Import the older project into the latest version of the Business Modeler IDE. This updates the data model to the latest data model version.
2. If your project contains custom code, you must treat it as though it were new code. Perform the following steps:
 - a. Clean up the **output** and **source** folders.
 - b. Generate Code for the project.
 - c. Perform these steps in stages to aid in troubleshooting:
 - A. Insert your old source code into the new files.
 - B. Build the project.
3. Package the template in the Business Modeler IDE.
4. Install the packaged template to the upgraded server.

Create a new Oracle tablespace

If you are upgrading from a Teamcenter version earlier than 11.2 and your environment uses Oracle, you must create a new, temporary tablespace before upgrading the corporate server. If this is not done, the upgrade fails. The new tablespace must be at least twice the size of the current **PIMANRELATION** table. This tablespace can be removed after the corporate server upgrade is complete.

You can create this table manually before launching TEM to perform the upgrade, or TEM can create the tablespace during the upgrade. If you prefer to create the tablespace manually, follow these steps. If you plan to have TEM create the tablespace, skip these steps.

1. Open a command prompt and log on to **sqlplus** as the Oracle user for the Teamcenter database, for example, **tcdba**.
2. Determine the size of the current **PIMANRELATION** table by entering:

```
WITH cte1 AS ( SELECT table_name FROM user_tables WHERE
table_name='PIMANRELATION' UNION ALL SELECT index_name FROM
user_indexes WHERE table_name='PIMANRELATION' UNION ALL SELECT
```

```
index_name FROM user_indexes WHERE table_name='PIMANRELATION'), cte2
AS ( SELECT SUM(blocks)*8/1024 MB from dba_extents, cte1 where
owner='Tc-Oracle-user' and segment_name=cte1.table_name) SELECT SUM(MB)
FROM cte2;
```

This query returns the size of the table in megabytes.

3. Log off **sqlplus** by entering: **exit**.
4. Log on to **sqlplus** as **system**.
5. Create a new tablespace with double the size of the value returned by the query. The tablespace name can be any unique value.

Oracle has a 32 GB limit on database file size. If you need more than 32 GBs for the temporary tablespace, you need to create multiple database files that are less than 32 GBs and assign them to the same tablespace. For example, if you need a temporary tablespace that is 100 GBs, create four database files.

Enter:

```
create tablespace tablespace-name datafile 'location/dbf-filename' size XM;
```

X is double the size of the current **PIMANRELATION** table returned by the query.

For example, if the query returned a value of 50 MBs and the tablespace name is **CSR**:

```
create tablespace CSR datafile 'D:\apps\oracleversion\oradata\tcversion\CSR.dbf' size 100M;
```

6. Grant all permissions on the new tablespace to the Teamcenter Oracle user:
 - a. Enter:

```
alter user Tc-Oracle-user quota XM on tablespace-name;
```

Where X is the size of the new tablespace.

For example:

```
alter user Tc-Oracle-user quota 100M on CSR;
```

- b. Enter:

```
grant unlimited tablespace to Tc-Oracle-user ;
```

7. Log off **sqlplus** by entering: **exit**.

8. During the corporate server upgrade, TEM must access the new tablespace; therefore, you must provide its name to TEM. If you do not set this environment variable, the upgrade will fail.

This is done by setting the `TC_TEMP_TABLESPACE_FOR_COPY_STABLE_ID` environment variable to the new tablespace name. For example:

```
TC_TEMP_TABLESPACE_FOR_COPY_STABLE_ID=CSR
```

Upgrade a test system

Create an upgrade testing environment

Methods for creating an upgrade testing environment

Before you upgrade Teamcenter production systems, upgrade a test system to ensure that the upgrade works successfully with the latest templates.

Upgrading a test database allows you to identify and correct upgrade problems and plan your production system upgrade accordingly. This can help prevent upgrade failures and minimize downtime when implementing the upgrade on your production system.

If you use Deployment Center, create an upgrade environment as described in the *Deployment Center Guide*.

If you use TEM, you can create a test environment two ways:

- **Copy a Teamcenter environment using TEM.**
- **Manually create a test environment.**

Warning:

Do not copy a Teamcenter environment to provide multiple servers in a production environment. You should only copy a Teamcenter environment for test purposes.

Copy a Teamcenter environment using TEM

Teamcenter Environment Manager (TEM) can copy an existing Teamcenter environment for upgrade testing.

1. Copy your Teamcenter database using the utilities and documentation provided by your database vendor.
2. Copy any volumes that contain any data you may want to access before or after the test upgrade.

It is not necessary to copy any volume data, but volume data not copied would not be available in the copied environment.

Note:

There are three possible scenarios for copying volume data:

- **Copying no data**

TEM creates empty volumes to represent the actual volumes. New data can be created in these volume, but no existing data would be available in the copied environment.

- **Copying all data**

TEM provides the option (in the **Volume Information** panel) to specify individual directories. If all volume directories are copied and made available to the destination copy system, they can be specified. If all volume data is copied, all data is available in the copied environment.

- **Copying partial data**

The option in TEM to specify individual volume directories does not require that *all* directories be specified. If you want, only some of the volume data can be copied and TEM will create empty directories for any volume directories not copied. Data not copied would not be available in the copied environment.

3. Launch TEM from the installation media for your current Teamcenter version.
4. Proceed to the **Welcome to Teamcenter** panel and select **Teamcenter**.
5. In the **Install / Upgrade Options** panel, select **Create environment for upgrade testing** and then click **Install**.
6. In the **Media Locations** panel, specify locations of Teamcenter installation media.

If your current Teamcenter version is a minor release:

- Enter the location of the major release software kit in the **Original Media** box.
- Note that the location of the software kit from which you launched TEM is shown in the **Update Location** list.

In the **Update Location** list, add the locations of any other Teamcenter minor releases or patches included in your production environment. You must include all releases and patches to ensure the test environment reflects your production environment.

TEM applies updates in the order you specify. If updates contain different versions of the same software component, the update closest to the bottom of the list takes precedence. To change the order in which updates are applied, select an update in the list and click **Shift Up** or **Shift Down**.

7. In the **Configuration** panel, type a configuration name for the test environment.
8. In the **Test Environment Location** panel, enter the installation directory for the test environment.

Note:

TEM does not prompt you to select features. Features are installed in the test configuration based on the data models installed in the database.

9. In the **File System Cache Service (FSC)** panel, type the required values for creating the FSC. The FSC must be a master FSC.

The copy process creates its own FSC to support the volumes.

10. In the **Operating System User** panel, type the user's password in the **Password** and **Confirm** boxes and then click **Next**.

11. In the **Foundation Database** panel:

- Enter the database information for the copy of the database you created in step 1.
- Enter the location in which to create the test environment's *TC_DATA* directory in the **Data Directory** box.

After you enter access information for the copied database, TEM determines which features are installed from the data models registered with the database. If any features are not included in the Teamcenter software kit, locate those features in the following step.

12. In the **Database Configuration** panel, examine the list of features found in the database. If any feature is missing a **Path** value, click **Browse** to locate those features.

You must locate all features in the list before you can continue.

13. On the **Volume Information** panel, determine where to create the volumes associated with the database for the test environment.

Volume Information

The following volumes were found in the database. Browse to any volumes which were copied from the production system. Empty directories will automatically be created in the default volume directory for any volumes where the path has been left blank.

Default volume directory:

Volume Name	Orig. Host	Copied Volume Path
volume	lm6s003	

Note:

- Ownership of *all* volumes is transferred to the target system as part of the copy process.
- By default, TEM creates volumes in the path specified in the **Default volume directory** path. If you want to specify a location for a given volume, enter the path in the **Copied Volume Path** box for the volume.

The **Copied Volume Path** value for a volume must be unique. (No other volume may use this location.)

- The **Copied Volume Path** value can point to an actual volume directory, a partial volume directory, or an empty directory. TEM creates whatever subdirectories are missing.
- The originating host name of the volume is used in the default volume location. This is done in the case two volumes originating from different servers use the same path.

14. In the **Teamcenter Administrative User** panel, type the user name and password and then click **Next**.
15. In the **Flex License Server** panel, enter access information for the Teamcenter license server.
16. In the **Confirmation** panel, review the information shown, and then click **Start** to begin creating the test environment.

Note:

If you install a rich client that points to the `TC_DATA` directory of a copied corporate server environment, set the `TC_SKIP_CLIENT_CACHE` variable to **TRUE** in the appropriate rich client launch file:

- Linux systems: `TC_ROOT/portal/start_portal`
- Windows systems: `TC_ROOT\portal\portal.bat`

Set this variable *before* you launch the rich client that points to the copied server environment.

If your original corporate server environment contains a rich client *before* you copy the environment, this step is not necessary to launch that rich client. However, if you add an additional rich client, you must set `TC_SKIP_CLIENT_CACHE` to **TRUE** before you launch the additional client.

Manually create a test environment

1. Export a database backup from your production environment using the utilities and documentation provided by your database vendor.
2. In your test environment, create folders for each volume in your production environment. If your production environment has multiple volume servers, you can create all test volumes on a single test server. This reduces the number of volume servers involved in testing.

The absolute paths to the volumes on the test server is not important, but the directory structure and names of the subfolders under each test volume must match those of the production volumes. The upgrade process fails if all volumes and their subfolders are not re-created in the test environment.

3. You must copy all contents of the `dba_xxxxxxx` subfolder under the default volume owned by the Teamcenter database user in the production environment to the corresponding volume subfolder in the test environment; otherwise, the upgrade process fails if this is not done.
4. Copy data needed for testing from the production environment to the corresponding volume subfolders in the test environment. For the upgrade to succeed, it is not necessary to copy any volume data, but volume data not copied is not available for testing after the upgrade.
5. If custom templates are deployed in your production environment, export those templates to packages and copy them to your test server. You must install these packages when installing the test corporate server.
6. On the test server, install a corporate server and a two-tier rich client to the same Teamcenter version as your production environment.

In the **Features** panel in TEM, select the same list of features installed in your production environment including your custom template packages and third-party custom template packages. To install a custom template package:

- a. In the **Features** panel, click **Browse**.

TEM displays an **Open** dialog box.

- b. In the **Open** dialog box, browse to the location on the test server where you copied the custom template package. Select the feature file (**feature_template-name.xml**) for your custom template, and then click **Select**.
 - c. In the **Features** panel, the custom template package is now listed, under **Extensions**, but the template package is not selected. Select the custom template package so that it is installed.
7. Once the test corporate server is installed, import your production database backup into your test database using the utilities and documentation provided by your database vendor.
8. Preferences are stored in the database. Because you imported the production database backup into your test database, you need to modify a few preferences:
 - a. From a Teamcenter command prompt, run:

```
preferences_manager -u=user -p=password -g=dba -mode=export -out_file=%TC_DATA%\preferences-file.xml -scope=SITE
```

user/password must be a Teamcenter user in the **dba** group.

preferences-file.xml is the file to which the preferences are exported.

- b. Open *preferences-file.xml* in a text editor.
 - c. Modify the following preferences:

- **Fms_BootStrap_Urls**

The host name and port must match those of your test corporate server.

- **Transient_Volume_Installation_Location**

The host name must match that of your test corporate server.

- **WEB_default_site_server**

If you are using the four-tier architecture in your test environment, the host name and port must match that of your test corporate server.

- **Default_Transient_Server**

The host name and port must match those of your test corporate server.

- **Default_Transient_Volume_Id**

This value should be modified to match the value of the transient volume id defined in the **backup_xmlinfo** result and the FMS master file.

- **Transient_Volume_RootDir**

The path must match that of the transient volume on your test corporate server. For example:

On Linux: **/tmp/transientVolume_infodba**

On Windows: **C:\Temp\transientVolume_infodba**

- d. Save and close *preferences-file*.
- e. In the Teamcenter command prompt, load the modified *preferences-file* into the database by running:

```
preferences_manager -u=user -p=password -g=dba -mode=import -file=%TC_DATA%\preferences-file.xml -scope=site -action=override
```

9. From a Teamcenter command prompt, regenerate the POM schema file by running:

```
install -regen_schema_file -u=Tc-db-user -p=password -g=dba
```

10. From a Teamcenter command prompt, regenerate the POM schema transmit file by running:

```
install -gen_xmit_file -u=Tc-db-user -p=password -g=dba
```

11. Modify the FMS master file:

- a. From a Teamcenter command prompt, run:

```
backup_xmlinfo -u=user -p=password -g=dba
```

user/password must be a Teamcenter user in the **dba** group.

This command generates a file in the directory from which it was run called **backup.xml**. This file contains the FMS information stored in the database. This includes the production environment's enterprise ID, volume IDs, and volume paths. You use information from **backup.xml** to update the test environment's FMS master file.

- b. Open **backup.xml** and the test environment's FMS master file in a text editor.
- c. In the FMS master file, make these changes:

- In the FMS master file's **fmsenterprise** tag, set the **id** attribute to the value of the **enterpriseID** tag in **backup.xml**.

This a 10-digit integer. Be sure to include the negative sign (-) at the beginning of this value.

- In the FMS master file, there is only one **volume** tag (excluding the transient volume). Set the **enterpriseid** attribute of that tag to the value of the **enterpriseID** tag in **backup.xml**.

This a 10-digit integer. Be sure to include the negative sign (-) at the beginning of this value.

- In the FMS master file, you must have one **volume** tag for each volume listed in **backup.xml**. In the FMS master file, copy the **volume** tag and paste as many volume tag copies as needed.
- For each volume listed in **backup.xml**, copy the value of the **volumeUid** tag and paste it in the **id** attribute of a **volume** tag in the FMS master file. Set the **root** attribute of that **volume** tag to the location of that volume on the test server.
- In the FMS master file, set the **enterpriseid** attribute of the **transientvolume** tag to the value of the **enterpriseID** tag in **backup.xml**.

This a 10-digit integer. Be sure to include the negative sign (-) at the beginning of this value.

- In **backup.xml**, there are two **transientVolumeInfo** tags, one for Windows and one for Linux. You must copy the appropriate **transVolId** value (Windows or Linux). Paste the copied value into the **id** attribute of the **transientvolume** tag in the FMS master file.

d. Save and close the FMS master file.

12. On the test server, stop and restart the FSC services.

13. On the test server, log on to rich client using an account in the **dba** group and open the Organization application.

14. For each volume:

- a. Set **Node Name** to the name of the test server on which the volume resides.
- b. Click **Modify**. Keep the volume panel open.
- c. Set either **Linux Path Name** or **Windows Path Name** to the volume's location on the test server.
- d. Click **Modify**.

- e. In the **Move volume** popup window, click **No**. (You are not trying to move files. You are only editing the path.)
 - f. In the second **Move volume** popup window, click **Yes**.
15. Run **index_verifier** to verify that all database indexes are present. Missing indexes can cause performance issues after the database is upgraded.
- a. Open a Teamcenter command prompt.
 - b. Optionally, perform a dry run to identify missing indexes but do not create missing indexes:

```
TC_ROOT\bin\index_verifier -u=Tc-db-user -p=password -g=dba -o=DRYRUN > missing-  
indexes.sql
```

missing-indexes.sql is the file to which index creation statements for missing indexes are written.

- c. Create missing indexes using one of these methods:
 - From the Teamcenter command prompt, run:


```
TC_ROOT\bin\index_verifier -u=Tc-db-user -p=password -g=dba -o=DRYRUN
```
 - Use a database vendor tool (such as SQL*Plus for Oracle, SQL Server Management Studio or **sqlcmd** for Microsoft SQL Server) to process the index creation statements in the *missing-indexes.sql* file generated by running **index_verifier** with the **DRYRUN** option.
16. Launch the rich client in your test environment to verify custom functionality.

Upgrade the test environment

Launch the installer

1. Locate the Teamcenter 14 and the Teamcenter 14.3 software kits.
2. If you use Teamcenter Integration for I-deas, run the **TcII.bat** script (on Windows systems) or the **TcII.sh** script (on Linux systems) on the Teamcenter Integration for I-deas software kit. This sets the **CLASSPATH** variable and other variables required for TEM to find installation files for the integration.
3. Specify the path to the Java Runtime Environment (JRE) in the **JRE64_HOME** environment variable on your host.
4. Start Teamcenter Environment Manager (TEM):

- a. Change to the root directory of the Teamcenter 14.3 software kit.
- b. Launch TEM:
 - Windows systems: Right-click the **tem.bat** program icon and choose **Run as administrator**.
 - Linux systems: Run the **tem.sh** script.
5. Proceed to the **Welcome to Teamcenter** panel and choose **Teamcenter**.
6. In the **Install / Upgrade Options** panel, click **Upgrade**.
7. In the **Media Locations** panel, specify locations of Teamcenter software kits:
 - a. In the **Original Media Location** box, enter the location of the Teamcenter 14 software kit.
 - b. The **Update Location** table automatically contains the location of the current minor release (Teamcenter 14.3). If you want to apply any additional patches during upgrade, click **Browse** to add the locations of patches to the **Update Location** table.

TEM applies updates in the order you specify. If updates contain different versions of the same software component, the update closest to the *bottom* of the list takes precedence. To change the order in which updates are applied, select an update in the list and click **Up** or **Down**.

Identify old and new configurations

1. In the **Old Application Root** panel, enter the path to *TC_ROOT* directory of the corporate server to be upgraded.
2. In the **Old Configuration** panel, select the corporate server configuration you want to upgrade.
3. In the **Configuration** panel, type an ID and a description for the new corporate server.
4. Proceed to the **New Application Root** panel.

This panel contains a list of features in your current installation and the status of each feature:

- A status of **Upgrade** means the feature will be upgraded to target Teamcenter release.
- A status of **Deprecated** means the feature will not be upgraded.
- A status of **WARNING: Browse to a feature XML file** indicates that you must provide the location of the upgraded custom feature file. Click **Browse**, navigate to the package that was exported from the upgraded custom Business Modeler IDE template project, and select the feature XML file. After you provide the location of the feature file, the status will be **Upgrade**.

In the **New Application Root Directory** box, enter the root directory for the new corporate server. This is the *TC_ROOT* directory for the upgraded Teamcenter installation.

5. In the **Operating System User** panel, type the password for the operating system user performing the upgrade.
6. In the **Teamcenter Administrative User** panel, type the password for the Teamcenter administrator account.
7. Proceed to the **Upgrade Information** panel and enter the required values.

Value	Description
Old TC_DATA Location	Specifies the path to the <i>TC_DATA</i> directory for the corporate server to be upgraded. Verify that the path shown is correct.
New TC_DATA Location	Specifies the path you want to use for the new <i>TC_DATA</i> directory on the upgraded corporate server.
Database User	Specifies the user name of the Teamcenter database user.
Database Password	Specifies the password for the database user.
Caution: The password must not be empty nor contain any whitespace characters such as space, tab, newline, carriage return, form feed, or vertical tab. In addition, the password must not contain any of the following characters: ! @ \$ % = & ' " ^ : ; . _ < > () { }	

8. The **Confirmation** window informs you that providing the correct information in the **Upgrade Database Features** panel is highly important. Failure to do so will result in issues with your upgrade and migration of custom data model definitions. Click **Confirm**.
9. The **Upgrade Database Features** panel contains a table of feature templates found in the database with the status of each in the **Status** column.

If TEM prompts you for a custom template, click the **Browse** button and navigate to the newly packaged template and libraries.

Standard Teamcenter features whose status is **Upgrade** are included in the Teamcenter software kit and will be upgraded to the target Teamcenter release.

If your site depends on templates from asynchronously released features or on templates supplied by other sites, partners, or third parties, click the **Browse** button to locate the template package (**ZIP**) file for each missing template.

You must provide paths to all feature templates before you can proceed with upgrade. The **Next** button is disabled until valid paths are provided for all templates.

Failure to locate the correct paths for all template packages results in migration issues with your custom data model to the Business Modeler IDE, possible corruption of data, and problems with the Teamcenter server.

Configure components

1. Complete the **Upgrade Options** panel.

The **Generate client cache** option specifies that you want to generate a cache of data that rich clients can download once at initial logon and then reuse on the client host. This option reduces server demand, reduces startup time, and improves overall performance. When this option is selected, TEM generates the client cache at the end of the install, upgrade, or update action. If you clear this option, but a client cache already exists, the old client cache is deleted.

2. In the **Flex License Client** panel, enter the license server and license file information.
3. In the **Password Security** panel, define an **Administrative Password Directory**. Encrypted password files are kept in this directory.
4. In the **TcServer Character Encoding Settings** panel, choose the appropriate **Canonical Name** for the character set the rich client uses to access the database.

Caution:

To prevent data corruption, this character encoding set must match the encoding set used by the Teamcenter database.

5. If the **Server Manager** feature is installed on the server being upgraded, the **Multiplexing Proxy (MUX)** panel appears.

The MUX listens on a single port for incoming requests from the web tier, forwards those requests to a server process, and streams responses back to web tier.

- **Port**

Specifies the TCP/IP port on which the MUX listens for web tier requests

- **TECS Admin Port**

Specifies the port used by the Teamcenter Enterprise Communication System (TECS).

6. If the **Server Manager** feature is installed on the server being upgraded, the **Communication Configuration** panel appears. It is prepopulated with the values from the existing server. Modify values as needed.
7. If the **Server Manager** feature is installed on the server being upgraded, the **Server Manager Cluster Configuration** panel appears. It is prepopulated with the values from the existing server. Modify values as needed.

Start the upgrade

1. Proceed to the **Pre-Upgrade Diagnostics** panel.

Enter the path to a directory in which to store upgrade diagnostic log files.

Click **Run** to begin pre-upgrade diagnostics.

TEM performs a series of diagnostics on the Teamcenter configuration and reports any problems found.

2. Allow time for the pre-upgrade diagnostic tests to complete.

If any test fails, click the appropriate **View** button to view details of the failed test. Click **Review** to view complete results of the pre-upgrade diagnostics.

Perform any recommended steps in the diagnostic reports to resolve the failures. After performing these steps, click **Run** to perform pre-diagnostic tests again. TEM does not allow the upgrade to proceed until pre-diagnostic failures are resolved.

If all diagnostic tests are successful (the **Results** column displays **Passed** for all tests), click **Next** to continue.

If your corporate server is hosted on a Linux machine, **C-Shell must be installed**, otherwise, the Pre-Upgrade Diagnostic test fails at the Volume Access Check.

3. In the **Confirmation** panel, review your selections. If you need to make a change, click **Back**. When you are ready to begin the upgrade, click **Start**.

The **Upgrade** panel displays the status of the upgrade.

4. If the upgrade is successful, close TEM.

If the upgrade is not successful, click **Navigate Logs** to open the upgrade log viewer.

Post-upgrade tasks

Run the postinstallation tasks script (Linux systems)

you installed the corporate server without root privileges, a user with root privileges must run the root postinstallation tasks script. This script registers daemons and performs other installation actions that require root privileges.

In the `TC_ROOT/install` directory, locate and run the following script:

```
root_post_tasks_id.ksh
```

Enable a UTF-8 database

If you converted your Teamcenter database from non-UTF-8 to UTF-8 before you upgraded Teamcenter, update the **uTF8Enabled** setting in the **configuration.xml** file:

1. Locate the `TC_ROOT\install\configuration.xml` file in your upgraded environment and open the file.
2. Locate the **uTF8Enabled** parameter in the file and set its value to **true**:

```
<uTF8Enabled value="true" />
```

3. Save the changes to the file.

Resolve Schedule Manager model errors

In current Teamcenter versions, the **Schedule**, **Schedule Task**, **Schedule Revision**, **Schedule Task Revision** and the associated forms (**Scheduling**, **Execution**, **SchMgtCostForm**) are re-parented from **Item** and **Item Revision** to **Workspace Object**.

In the default template, all references from **Schedule** and associated objects to **Item** and **Item Revision** have been corrected. However, for custom templates containing Schedule Manager customizations, the following steps must be performed to resolve any model errors during the Business Modeler IDE template migration.

Associated forms (**Scheduling**, **Execution**, **SchMgtCostForm**), **Schedule Revision** and **ScheduleTaskRevision** are not used by Schedule Manager and will be deprecated in a future release of Teamcenter.

Examine model errors during migration of custom template to Teamcenter 14.3

Study the model errors reported in the console of Business Modeler IDE during migration of the Business Modeler IDE custom template. These errors are typically in the **extensions/default.xml** and **extensions/lang/en_US/default_en_US.xml** files. Look for errors similar to the following examples:

- Property "pv5_MFK_Prop1" is not attached to Business Object "PV5_MFK_SchdTsk1" in TcTypeConstantAttach element.
- Invalid type found in constant attach element. "PV5_MFK_SchdTsk1" is not a subtype of any constant scope definition
- Incompatible MFK Definition: The property named "pv5_MFK_Prop1" on MFK definition "PV5_MFK_Require" is not valid on the business object "PV5_MFK_SchdTsk1".
- The OperationInputType "PV5_MFK_SchdTsk1RevisionRevI" cannot be loaded because the parent Type "ScheduleTaskRevisionRevI" does not exist.
- Property "pv5_MFK_Prop1" is not attached to Business Object "PV5_MFK_SchdTsk1" in TcPropertyConstantAttach element.
- Property "items_tag" is not attached to Business Object "Mr9a_SSJbTsk1Revision" in pathToSource element.
- Invalid type found in constant attach element. "Mr9a_SSJbTsk1" is not a subtype of any constant scope definition.
- Property "mr9a_jobact2prop" is not attached to Business Object "Mr9a_SSJbTsk1Revision" in Property{::}Mr9a_SSJbTsk1Revision{::}mr9a_jobact2prop element
- Attaching a Naming Rule to the "item_id" property on the "ScheduleTask" Business Object is not supported.

Resolve the model errors

1. Open the XML file in a text editor, XML editor, or the Business Modeler IDE text editor.
2. Find the line number where the error is reported and delete the invalid element at that line number in the XML file.
3. Save the XML file.
4. Reload the custom template in BMIDE again and ensure there are no model errors being reported in the custom template.

Caution:

Do not remove **TcStandardType** and **TcClass** model statements for any subtypes of **Schedule** and **ScheduleTask** business objects.

After the upgrade process in TEM is complete, the custom **Schedule** and **ScheduleTask** business objects are automatically re-parented to **WorkspaceObject**.

Upgrade Access Manager rules

Special steps are required to upgrade the Access Manager rule tree. These steps are required to ensure the rule tree in your upgraded system contains any new rules added by Teamcenter and also any custom rules you added to your previous installation. You can upgrade rules in two ways. Choose the appropriate method depending on how many custom rules are in your Access Manager rule tree.

- If you have many custom rules, migrate your legacy rule tree and then manually add new Teamcenter-supplied rules:
 1. Create a backup of your existing rule tree using the Access Manager **Export** action.
 2. Upgrade your Teamcenter configuration to Teamcenter 14.3. During Teamcenter upgrade, TEM automatically imports your legacy rule tree to Teamcenter 14.3.
 3. Identify changes in the rule tree by comparing the `..ITC_DATA\tc_am_rule_tree.default` file in your previous environment to the same file in your Teamcenter 14.3 environment.
 4. Start Access Manager and add rules introduced to Teamcenter since your previous version.
- If you have few or no custom rules, use the standard Teamcenter 14.3 rule tree and then manually add your custom rules.
 1. Create a backup of your existing rule tree using the Access Manager **Export** action.
 2. Identify your custom rules in order to add them after upgrade.
 3. Upgrade your Teamcenter configuration to Teamcenter 14.3. During Teamcenter upgrade, TEM automatically imports your legacy rule tree to Teamcenter 14.3.
 4. Import the standard Teamcenter 14.3 rule tree using the **am_install_tree** utility. Use the **mode=replace_all** argument to overwrite the legacy rule tree with the Teamcenter 14.3 rule tree. The utility automatically creates ACLs and privileges during import.
 5. Manually add your custom rules into the rule tree in the appropriate locations.

The Access Manager supports localization. This includes locale-specific display names of access control list (ACL) objects, privilege names, and accessor type values such as group names and role names. This localization capability is provided using text server XML files. The rule tree import/export functionality supports XML format input files.

The **am_install_tree** utility supports both ASCII text format and XML format rule tree files. However, export in the Access Manager application generates the output file only in XML format. This allows exported ACL name translations to be migrated to other sites.

An XML Access Manager rule tree resembles the following example.

```
<?xml version="1.0" encoding="UTF-8"?>
<Tc_data_access_config>
  <privileges>
    <priv_name>READ</priv_name>
    <priv_name>WRITE</priv_name>
    <priv_name>COPY</priv_name>
    <priv_name>CHANGE</priv_name>
    <priv_name>DELETE</priv_name>
  </privileges>

  <named_acls>
    <named_acl>
      <acl_name>Working</acl_name>
      <acl_name language="fr_FR">working_fr</acl_name>
      <acl_name language="de_DE">working_de</acl_name>
      <acl_name language="jp_JP">working_jp</acl_name>
      <ace_entry>
        <accessor_type>group</accessor_type>
        <accessor>dba</accessor>
        <grant>
          <p>READ</p>
          <p>WRITE</p>
          <p>COPY</p>
        </grant>
        <revoke>
          <p>DELETE</p>
          <p>CHANGE</p>
        </revoke>
      </ace_entry>
      <ace_entry>
        <accessor_type>Owning Group</accessor_type>
        <accessor> </accessor>
        <grant>
          <p>READ</p>
          <p>WRITE</p>
          <p>COPY</p>
        </grant>
        <revoke>
          <p>DELETE</p>
          <p>CHANGE</p>
        </revoke>
      </ace_entry>
    </named_acl>
    <named_acl>
      <acl_name>In Project ACL</acl_name>
      <acl_name language="fr_FR">In Project ACL fr</acl_name>
      <acl_name language="de_DE">In Project ACL de</acl_name>
      <acl_name language="jp_JP">In Project ACL jp</acl_name>
      <ace_entry>
        <accessor_type>group</accessor_type>
        <accessor>dba</accessor>
```

Access Manager rule tree example (Continued)

```

<grant>
  <p>READ</p>
  <p>WRITE</p>
  <p>COPY</p>
</grant>
  <revoke>
    <p>DELETE</p>
    <p>CHANGE</p>
  </revoke>
</ace_entry>
<ace_entry>
  <accessor_type>Owning Group</accessor_type>
  <accessor> </accessor>
  <grant>
    <p>READ</p>
    <p>WRITE</p>
    <p>COPY</p>
  </grant>
  <revoke>
    <p>DELETE</p>
    <p>CHANGE</p>
  </revoke>
</ace_entry>
</named_acl>
</named_acls>
<rule_tree>
  <tree_node>
    <rule_name>Has Class</rule_name>
    <rule_argument>POM_object</rule_argument>
    <acl_name></acl_name>
    <tree_node>
      <rule_name>Has Bypass</rule_name>
      <rule_argument>true</rule_argument>
      <acl_name>Bypass</acl_name>
    </tree_node>
    <tree_node>
      <rule_name>Has Status</rule_name>
      <rule_argument></rule_argument>
      <acl_name>Vault</acl_name>
    </tree_node>
    <tree_node>
      <rule_name>Has Class</rule_name>
      <rule_argument>POM_application_object</rule_argument>
      <acl_name>Working</acl_name>
      <tree_node>
        <rule_name>Has Class</rule_name>
        <rule_argument>Dataset</rule_argument>
        <acl_name>Dataset ACL</acl_name>
      </tree_node>
    </tree_node>
  </tree_node>
</rule_tree>
</Tc_data_access_config>

```

Access Manager rule tree example

Maintain default scope rules

Default scope rules (transfer modes, closure rules, filter rules, and so on) are maintained in XML files in the `TC_DATA` directory (for example, **defaultTransfermodes.xml**). The file is imported to Teamcenter using the **tcxml_import** command line utility during the database installation and upgrade.

The transfer mode **.xml** files are imported in overwrite mode as part of the upgrade. You need not manually import the files. To reload the latest changes from the **.xml** file, you can run the **tcxml_import** utility in overwrite mode to ensure that the updates to existing scope rules are also imported to the database. For example:

```
$TC_ROOT/bin/tcxml_import -u=user -p=password -g=group -file=$TC_DATA/
defaultTransfermodes.xml -scope_rules -scope_rules_mode=overwrite
```

Siemens Digital Industries Software recommends that you do not modify the default scope rules. If necessary, you can create a copy of the default rules and modify the copy as needed.

Caution:

All references to obsolete attributes and classes must be removed from custom closure rules or property sets. Additionally, remove any deprecated attributes and classes from custom closure rules or property sets.

For information about deprecated and obsolete classes and attributes, see the latest Teamcenter **README** file in the **Downloads** area on Support Center.

Set the time zone

Some Teamcenter applications including Schedule Manager and Workflow utilize calendars. When creating new calendars, Teamcenter determines the default time zone as follows:

1. Teamcenter checks the **Time Zone** property on the default calendar. If this property is set, its value is used.
2. If that property is not set, Teamcenter checks the **SiteTimeZone** preference. If this preference is set, its value is used.
3. If neither of these are set, Teamcenter uses GMT as the time zone.

If Teamcenter is using GMT and that is not the correct time zone for your site, you may encounter unexpected behavior. Therefore, Siemens Digital Industries Software recommends that you set both the **Time Zone** property on the default calendar using the Organization application and the **SiteTimeZone** preference.

Additional postupgrade steps

- **Transfer access rules for Classification Admin objects**

If you created classification access rules for admin objects that resided in the **Has Class(POM_object)** area, you must move these rules manually to the **Has Application(Any)** area of the rule tree after upgrade.

In current versions of Teamcenter, access to classification administrative objects is restricted to dba users, but you can grant other users access to classification objects by creating a rule for each of these users or groups of users in the **Has Application(Any)** group in the rule tree.

- **Add new Classification units**

When you upgrade Teamcenter, new Classification units are not added because they may conflict with other units created by Teamcenter administrators. Update unit definitions manually as described in *Classification Admin*.

- **Set variant rule preference**

If you already use classic variants, you can use your existing variability data with the Product Configurator. You can now use variant rules to store variant configuration criteria and optional validation records. They are attached to the product item revision or the application model (collaborative design) by GRM relationships.

Variant rules perform the same function as stored option sets (SOSs), which will be deprecated in a future version. Siemens Digital Industries Software encourages customers who use classic variants to transition to variant rules by setting the **DisableSaveSos** preference to **TRUE**. This setting prevents users from saving SOSs, thereby forcing them to create variant rules.

Existing modular variant customers should not change the default preference setting of **false**.

- **Transfer HTTPS certificates**

If you use HTTPS transports, you must transfer all the HTTPS certificate information to the upgraded Teamcenter installation.

7. Update Active Workspace and microservices

If you are running Active Workspace 4.2 or earlier, you must add the microservice framework before you update to Active Workspace 6.3, as described in *Add microservice framework and update Active Workspace*.

If your Teamcenter environment does not yet include Active Workspace, add microservice framework and Active Workspace as described in *Add Active Workspace to a Teamcenter environment*.

If you are running Active Workspace 4.3 or later, your environment includes the microservice framework. Update the microservice framework and Active Workspace to version 6.3 by performing the following steps:

1. Update your Teamcenter environment (patch or upgrade) to a version supported for Active Workspace 6.3.

Make sure you update every server in your environment.

2. **Patch microservice framework.**

Make sure you patch every server in your environment.

On Linux microservice nodes, to ensure new Docker images are used, redeploy them using the updated **.yml** files.

3. Update Active Workspace:

- a. **Patch Active Workspace.**

Make sure you patch every server in your environment.

- b. **Update the Active Workspace environment.**

- c. **Publish changes to the file repository.**

Patch Active Workspace 4.3 or later

Apply Active Workspace patches on *all* Teamcenter hosts that have Teamcenter Foundation installed.

Apply microservice framework patches on hosts where the microservice framework is installed.

1. Make sure you have the correct **Active Workspace and microservice framework software kits**.

2. Stop all Teamcenter services *except* FSC services. Make sure all server manager services are stopped.
3. Launch TEM in maintenance mode for the configuration you are updating.

On Windows systems, launch TEM with administrative privileges (right-click the **tem.bat** icon→**Run as administrator**).

4. In the **Maintenance** panel, select **Updates Manager**.
5. Proceed to the **Apply Updates** panel and enter required paths:
 - In the **Update kit location** box, enter the path to the expanded microservice framework software kit.
 - In the **Backup directory** box, enter a path in which you want TEM to create backups of files replaced during patching.
 - In the **Original Media Location** box, enter the path to the software kit for the major version of Teamcenter or Teamcenter Rapid Start.
6. Proceed to the **Operating System User** panel and type the password of the operating system account used to update Teamcenter.
7. Proceed to the **Teamcenter Administrative User** panel and type the user's password.
8. Proceed to the **Confirmation** panel and click **Start** to apply the patch that adds Active Workspace to your environment.
9. When the update is complete, close TEM.
10. Repeat steps 3 through 9 to apply the Active Workspace 6.3 patch.
11. On hosts with Teamcenter Foundation installed, update the *TC_DATA* directory:
 - a. In the location of the expanded Active Workspace 6.3 software kit, browse to the **tc** directory.
 - b. Expand the **data.zip** file from this location to the *TC_DATA* folder.

If you want to add additional Active Workspace applications and components to your environment, restart TEM in maintenance mode and select additional Active Workspace Server Extensions, client features, or microservices in the **Features** panel.

Continue updating Active Workspace by **updating the Active Workspace environment**, and then **publishing changes to the file repository**.

Complete the Active Workspace update

Remerge Solr and Teamcenter schemas and update the index

After moving to the target Active Workspace and Teamcenter versions, the Teamcenter and Solr schemas are not synchronized. You can merge the schemas and then choose the method for updating the index. You must determine if you need a full index or delta reindex of your data before planning your index update.

Evaluate the delta of object data changes

If your indexing changes are additions, modifications, and deletions for types and properties, you can perform a delta indexing update rather than a complete reindex.

1. Merge the Teamcenter and Solr schemas.
2. Stop synchronization by the indexer if it's running.

```
runTcFTSIndexer -stop
```

3. Determine the scope of the changes between the last indexing schema and the current schema.

Run the **awindexerutil** utility using **-delta -dryrun** to get a report of the expected delta of changes. For example:

```
awindexerutil -u=adminuser -p=password -g=group -delta -dryrun
```

The differences are output to the command window as well as to a log file.

4. After you evaluate the report, determine whether you want to use the delta of changes for reindexing. If so, run **awindexerutil** to index the changes from the report:

```
awindexerutil u=adminuser -p=password -g=group -delta
```

5. Test indexer connectivity by running the indexer test flow.

```
runTcFTSIndexer -task=objdata:test
```

6. Restart the synchronization flow using the **runTcFTSIndexer** utility:

```
runTcFTSIndexer -task=objdata:sync -interval=seconds
```

Reindex your data if needed

Your index may have a high number of changes or other kinds of changes that are outside the scope of delta changes. If that is the case, reindex your data instead:

1. Merge the Teamcenter and Solr schemas.
2. Test indexer connectivity by running the indexer test flow.

```
runTcFTSIndexer -task=objdata:test
```

3. Reindex the data.

Additional postupdate steps

Import table column configuration

If you have configured any custom table columns other than those available by default, perform the following after upgrading or patching:

1. Update the custom column information in the **Awb0ContentTableUiConfigCots.xml** file.
2. Run the **import_uiconfig** utility to import the column configuration.

Reindex classification data

If you are new to classification and want to search for classification data in Active Workspace using classification standard taxonomy, your first step is to create a classification hierarchy.

If you already use traditional classification and want to search for classification data in Active Workspace or Teamcenter Manufacturing Access, perform the following steps:

1. Create search index views and specify facetable properties.
2. Update and merge the schema file.
3. Index or reindex your data. If your data is already indexed, it is sufficient to index only the modified classification data.

These steps are described in *Classification and Library Management*

Install Systems Modeling and Systems Modeler features

Starting in Active Workspace 3.0, some functionality in the **Architecture** toolbar of the Client interface has been modularized. This functionality includes the **Start Authoring** in the **Architecture** toolbar. To restore this functionality after moving to Active Workspace 6.3:

- Install the **Systems Modeling** Server Extensions feature on the corporate server.
- Install the **Systems Modeler** Client feature in the Active Workspace client.

Set up a custom template after patching

If you have a custom template that is dependent on the Active Workspace template (**aws2**) and other Active Workspace templates, you must follow a special procedure to set up this template after patching Active Workspace.

1. Add the Active Workspace template (**aws2**) any other needed Active Workspace templates to the Business Modeler IDE by selecting **Add/Update Templates for working within the Business Modeler IDE Client** in the **Feature Maintenance** panel of Teamcenter Environment Manager (TEM).
2. Run TEM and re-patch Active Workspace.

This copies the required Business Modeler IDE .jar files for Active Workspace into the installation.

3. In the Business Modeler IDE, import your custom template. To add the Active Workspace template (**aws2**) and the other dependent Active Workspace templates:
 - a. Right-click your custom template project and choose **Properties**.
 - b. In the left pane, choose **Teamcenter**→**BMIDE**.
 - c. In the **Dependent Templates** pane, select the **aws2** template and the other Active Workspace templates previously added using the **Add/Update Templates for working within the Business Modeler IDE Client** option in TEM.
 - d. Click **OK**.

The template is now ready to be deployed.

Restart Teamcenter Process Manager

After completing updates to Active Workspace, ensure the updates made to the Teamcenter GraphQL Service take effect:

1. Stop the Teamcenter Process Manager service in the **Services** dialog of the Windows Control Panel.
2. Wait for all affected processes to stop, at least 15 seconds.
3. Restart the Teamcenter Process Manager service.

Some Active Workspace features such as Discussions (Active Collaboration) and Reports rely on updates to the Teamcenter GraphQL Service in order to function properly. This restart process ensures the updates are implemented for those features.

Publishing changes to the Active Workspace file repository

After patching, upgrading, or adding features to the Active Workspace client, publish changes to the Active Workspace file repository.

1. Start TEM on the Active Workspace Gateway host.
2. In the **Maintenance** panel, select **Configuration Manager**.
3. In the **Configuration Maintenance** panel, select **Perform maintenance on an existing configuration**.
4. In the **Old Configuration** panel, select the configuration.
5. In the **Feature Maintenance** panel, select **Update Active Workspace client settings**.
6. In the **Active Workspace Client Settings** panel, type the URL to the Active Workspace Gateway and enable publishing of client content to the Gateway.

Value	Description
Publish to Gateway	Select this check box to enable automatic publishing of Active Workspace content to the Gateway.
Gateway URL	Type the URL to Active Workspace Gateway: http://host:port Replace <i>host</i> with the host on which you installed the Gateway. Replace <i>port</i> with the port used by the Gateway. The default port is 3000 .

7. In the **Confirmation** panel, click **Start**.
8. When the installation is complete, close TEM.

8. Adding Active Workspace and microservices

If you are running Active Workspace 4.2 or earlier, you must add the microservice framework before you update to Active Workspace 6.3, as described in *Add microservice framework and update Active Workspace*.

If your Teamcenter environment does not yet include Active Workspace, add microservice framework and Active Workspace as described in *Add Active Workspace to a Teamcenter environment*.

Add microservice framework and update Active Workspace

Active Workspace architecture changed in Active Workspace 4.3, replacing the Active Workspace WAR file and .NET client with a system of microservices and an Active Workspace Gateway.

If you are running Active Workspace 4.2 or earlier, you must install microservice framework before you update to Active Workspace 6.3.

1. Update your Teamcenter environment (patch or upgrade) to a version supported for Active Workspace 6.3.

Make sure you update every server in your environment.

2. Install and configure the microservice framework on a Windows or Linux system. This includes:
 - Installing the microservice framework.
 - Installing the microservice proxy (microservice support for the web tier).
 - Starting the microservice node and any additional configuration for your platform type.
3. Update Active Workspace:
 - a. **Update the Active Workspace environment.**
 - b. **Publish changes to the file repository.**

Add Active Workspace to a Teamcenter environment

If you are adding Active Workspace to an existing Teamcenter environment, you must install the microservice framework before you install Active Workspace:

1. Update your Teamcenter environment (patch or upgrade) to a version supported for Active Workspace 6.3.

Make sure you update every server in your environment.

For supported combinations of Teamcenter or Teamcenter Rapid Start with Active Workspace, see the Hardware and Software Certifications knowledge base article on Support Center.

2. Install and configure the microservice framework on a Windows or Linux system. This includes:
 - Installing the microservice framework.
 - Installing the microservice proxy (microservice support for the web tier).
 - Starting the microservice node and any additional configuration for your platform type.
3. Install Active Workspace. This includes:
 - Active Workspace Server Extensions
 - Active Workspace Client, using Deployment Center or TEM (Windows or Linux)
 - Indexing, using Deployment Center or TEM (Windows or Linux)
 - Visualization Server, using Deployment Center or TEM (Windows or Linux)
 - Other components, according to your business needs

Patch Active Workspace 4.2 or earlier

Make sure you install and configure the microservice framework (on Windows or Linux) before you perform the following steps.

Apply the Active Workspace patch on *all* Teamcenter hosts that have Teamcenter Foundation installed.

1. Make sure you have the correct **Active Workspace software kit**.
2. If any server managers are running on the corporate server, shut them down before proceeding with the patch.
3. Remove your previous Active Workspace web application. Perform the appropriate step for the architecture you use:
 - Java EE: Remove the **awc.war** file and the **awc** directory under the **webapps** directory.
 - .NET: Stop the Teamcenter Active Workspace web site in Internet Information Server (IIS). Then, remove the Active Workspace web site.

Note:

The Active Workspace web application is not used in the current Active Workspace architecture.

4. Launch TEM in maintenance mode for the server you are updating.

5. In the **Maintenance** panel, select **Updates Manager**.
6. Complete the **Apply Updates** panel:
 - a. In the **Update kit location** box, enter the location of the Active Workspace 6.3 software kit.
 - b. In the **Backup directory** box, enter the location in which you want backup files to be created.
 - c. In the **Original Media Location** box, enter the location of the Teamcenter major release software kit, for example, Teamcenter 14.
7. In the **Operating System User** panel, type the required password.
8. In the **Teamcenter Administrative User** panel, type the required password.
9. If the patch contains enhancements to installed features in your configuration, TEM lists the enhancements in the **Required Configuration Enhancements** panel.

Features containing data model objects may be updated, meaning database model changes may occur. Siemens Digital Industries Software recommends you back up your database before continuing.

Select the **Accept Data Model Updates** check box to continue with installation.

10. If the server you are patching contains the Indexing Engine, TEM displays the **Indexing Engine Configuration** panel.

SOLR schema files location

- If the Indexing Engine and the Server Extensions are installed on the same server, the **SOLR schema files location** box is prepopulated with the correct location.
- If the Indexing Engine and the Server Extensions are installed on separate servers and you have already patched the server on which the Server Extensions are installed, enter the path to the upgraded Server Extensions server that contains the Solr schema files, `TC_DATA\fts\solr_schema_files`.
- If the Indexing Engine and the Server Extensions are installed on separate servers and you have not yet patched the server on which the Server Extensions are installed, leave the **SOLR schema files location** box blank. You can patch the Indexing Engine server now, but the schemas are not automatically merged. You must manually merge them after upgrading both the Indexing Engine server and the Server Extensions server.

Install indexing engine as a service

Select this to install the Indexing Engine as a service. If this is not selected, you must start the Indexing Engine manually.

11. In the **Publish to Gateway** panel, deselect the **Publish to Gateway** check box.

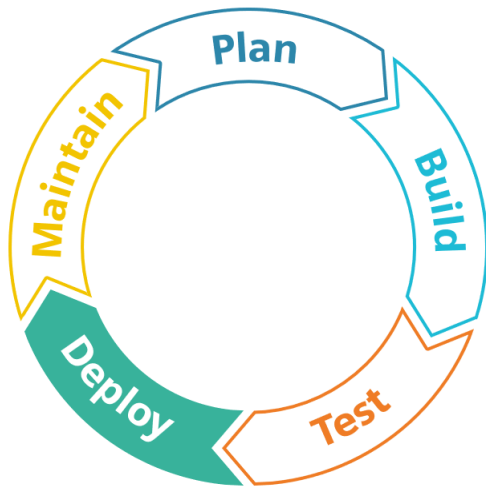
Publishing to Active Workspace Gateway cannot be enabled until after Active Workspace microservices are installed and running.

12. In the **Database User** panel, enter the database administrator password.
13. In the **Diagnostics** panel, enter a directory for storing log files and verify that all pre-patching tests are successful.
14. In the **Confirmation** panel, click **Start**.
15. During the update, a **Status Message** popup window opens. It informs you that you must manually patch the *TC_DATA* directory after TEM completes its updates.
16. When the update is complete, close TEM.
17. In the location of the expanded Active Workspace 6.3 software kit, browse to the **tc** directory.
18. Expand the **data.zip** file from this location to the *TC_DATA* folder on the patched host.

If you want to add additional Active Workspace applications and components to your environment, restart TEM in maintenance mode and select additional Active Workspace Server Extensions, client features, or microservices in the **Features** panel.

Continue updating Active Workspace by **updating the Active Workspace environment**, and then **publishing changes to the file repository**.

Part IV: Deploy the Teamcenter upgrade



After you successfully upgrade a test environment, deploy the upgrade to your production system by upgrading your volume servers and production environment.

9. Upgrade volume servers and the production environment

Upgrade volume servers

If you have volume servers (Teamcenter configurations that contain an FSC separate from the corporate server), you must upgrade those configurations *before* you upgrade the corporate server.

This is because Teamcenter utilities that run during the corporate server upgrade are not compatible with FSCs from other Teamcenter versions. TEM upgrades the FSC in the corporate server configuration during upgrade of the corporate server.

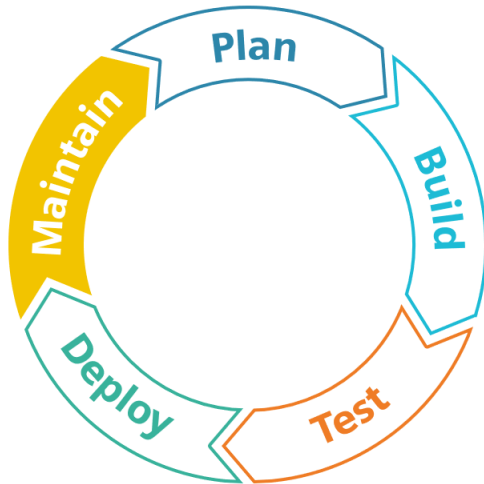
Upgrade the production environment

1. Open TEM in the new version and upgrade the production database.

During the upgrade, TEM automatically installs the dependent templates from the new version.

2. When TEM prompts for the custom template, click the **Browse** button and navigate to the newly packaged template and libraries.

Part V: Maintain Teamcenter between upgrades



If you use Deployment Center, apply patches to your environment as described in the *Deployment Center Guide*.

If you maintain your environment using Teamcenter Environment Manager (TEM), you apply software patches to your environment using the Updates Manager, an option in TEM when run in maintenance mode.

Patches to the Teamcenter Java EE web tier you install using the Web Application Manager.

Teamcenter updates (minor releases and patches) are posted for download on Support Center when available:

<https://support.sw.siemens.com>

10. Installing Teamcenter patches

Distribution of Teamcenter updates

Teamcenter updates are delivered in patches and in minor releases. Patches to Teamcenter servers and two-tier rich clients are applied using the Updates Manager in Teamcenter Environment Manager (TEM). Patches to the Java EE web tier are applied using the Web Application Manager.

Teamcenter patches are named using the following convention:

- *product-level_number_platform.zip*

Contains the server, TEM rich client, and web tier patches

- *product-level_number_PV_all.zip*

Contains the Teamcenter Visualization patches

- *product-level_number_install.zip*

Contains the TEM updates

Downloading Teamcenter patches

1. Locate the patches you want to apply in software downloads area on Support Center.
2. Download the patch files to a temporary location on your host.

Make sure you download the appropriate patch file for your platform.

Before you apply downloaded Teamcenter patches to your server, you must do the following:

- Install the base Teamcenter release.
- Stop all Teamcenter related processes and services (such as FSC, database daemons, and others).

TEM does not allow you to install patches for a version of Teamcenter that is earlier than your current installation. For example, you cannot apply a Teamcenter 12.3 patch to a Teamcenter 14.3 installation.

Patch Teamcenter Environment Manager

If the patch you downloaded has a corresponding installer patch file (*patch-id_install.zip*), download this installer patch and update Teamcenter Environment Manager (TEM) to the latest version using the following steps.

If no corresponding installer patch is posted on Support Center, use the existing TEM in your **install** directory under *TC_ROOT* and skip the following steps.

1. Copy the *patch-id_install.zip* file to your **install** directory under *TC_ROOT*.
2. Open a command prompt.
3. Change to the **install** directory under *TC_ROOT*.
4. Enter the following command to expand the *patch-id_install.zip* file, overwriting existing files:

```
7za x patch-id_install.zip -aoa -bb2 -bd
```

If errors occur while expanding the file, do one of the following tasks:

- Add the path to your *TC_ROOT\install\install* directory (on Windows systems) or *TC_ROOT/install/install* directory (on Linux systems) to your **PATH** environment variable and enter the **7za** command again.
- Enter the **7za** command with the full path to your *TC_ROOT\install\install* directory (on Windows systems) or *TC_ROOT/install/install* directory (on Linux systems). For example:

Windows systems:

```
TC_ROOT\install\install\7za x patch-id_install.zip -aoa -bb2 -bd
```

Linux systems:

```
TC_ROOT/install/install/7za x patch-id_install.zip -aoa -bb2 -bd
```

5. On Linux systems, add execute permissions back to the **tem.sh** shell script with the following command:

```
chmod 755 tem.sh
```

Migrate trace links

Beginning with Teamcenter 10.1.2.x patches, trace links are created on revisions of the absolute occurrence objects. In earlier versions of Teamcenter, trace links are created on absolute occurrence objects.

If your database contains trace links on absolute occurrence objects, run the **req_migrate_bomview_tracelinks** utility before you patch the corporate server. This utility migrates trace links on absolute occurrence objects to create trace links on the latest revision of the absolute occurrence objects.

You must have modify privileges on all the existing trace links in the database to run the migration. If you do not have modify privileges on some of the trace links, those trace links are not migrated, and a message is written to the log file.

1. In the Teamcenter environment, open a Teamcenter command prompt.
2. Type the following command:

```
req_migrate_bomview_tracelinks -u=Tc-admin-user -p=Tc-admin-password
-g=dba
```

The utility returns the number of trace links found in the database and lists each trace link as it is processed. The migration is complete when the command prompt is returned.

Estimate tablespace requirements for upgrade

Upgrading to Teamcenter 14.3 temporarily requires additional database tablespace.

To improve system performance, Teamcenter converts **POM_Stub** and **ImanExportRecord** objects to lightweight objects (LWOs). This changes the persisted schema, and these schema changes require some of the persisted data (including the **POM_Object** class table) to be reorganized. For large databases, these changes could take many days to complete by the prior approach of updating each row of the table. The Teamcenter upgrade process makes a copy of the table, inserting and modifying rows into the copy during the copying process. This upgrade process then replaces the original data tables with the modified data tables. This upgrade performance improvement requires more disk space, but only for the duration of the upgrade process.

Make sure your system has sufficient disk space to accommodate these needs during the upgrade process.

Oracle databases:

1. **Back up existing Teamcenter data.**
2. Determine the tables affected by the upgrade to Teamcenter 14.3.
 - a. Enter the following SQL command:

```
SELECT pname FROM PPOM_CLASS
WHERE BITAND(pproperties, 8192) <> 0
UNION
SELECT pname FROM PPOM_CLASS
WHERE BITAND(pproperties, 32768) <> 0;
```

- b. To the resulting list of tables, add the **PIMANEXPORTRECORD** and **PPOM_STUB** tables, which are also affected by the upgrade process.

- Determine the amount of space consumed by each affected table.

For each table identified in step 2 (including **PIMANEXPORTRECORD** and **PPOM_STUB**), enter the following SQL command:

```
WITH cte1
  AS (SELECT table_name
      FROM   user_tables
      WHERE  table_name = 'TABLE-NAME'),
  cte2
  AS (SELECT SUM(blocks) * 8 / 1024 MB
      FROM   dba_extents,
            cte1

WHERE  owner = (SELECT SYS_CONTEXT('userenv', 'current_schema')
                FROM   dual)
        AND segment_name = cte1.table_name)
SELECT ROUND(SUM(mb), 0) || ' MB' AS ESTIMATED_SPACE
FROM   cte2;
```

Replace *TABLE-NAME* with the name of the given table.

- Determine the amount of additional free space you need to complete the upgrade to Teamcenter 14.3.

Table	Space needed	Where space is needed
Tables found in step 2a	1.5 times current space	Current tablespace of the given table
PIMANEXPORTRECORD	1.5 times current space	Database user's default tablespace
PPOM_STUB	1.5 times current space	Database user's default tablespace

- Optionally, specify locations for new tablespaces and indexes.
 - If you want to ensure tables are rebuilt in a specific tablespace, set the **TC_UPGRADE_TABLE_STORAGE** environment variable to the appropriate tablespace name before you launch TEM to begin the upgrade.
 - If you want to ensure indexes for the upgraded tables are rebuilt in a specific tablespace, set the **TC_UPGRADE_INDEX_STORAGE** environment variable to the appropriate tablespace name before you launch TEM to begin the upgrade.

This applies *only* to tables identified in step 2a. The **PIMANEXPORTRECORD** and **PPOM_STUB** tables can only be rebuilt in the default tablespace.

After the upgrade is complete, you can move tables back to their original tablespaces.

Microsoft SQL Server databases:

1. **Back up existing Teamcenter data.**
2. Change the recovery model of the SQL Server database to **SIMPLE**.

After the upgrade, set the recovery model back to its previous value.

For information about changing the recovery model, see Microsoft SQL Server help:

<https://msdn.microsoft.com>

Perform a test upgrade. If the test upgrade fails during database upgrade, increase the available disk space for tablespaces and attempt the test upgrade again.

Install patches on a Teamcenter server

Caution:

If the patch contains an updated installer file (**install.zip**), **patch Teamcenter Environment Manager (TEM)** before you begin the following steps.

1. Expand the *product-level_number_platform.zip* file to a temporary location.
2. Shut down services:

- **Windows systems:**

If you use the .NET web tier, shut down Microsoft Internet Information Services (IIS) before you begin installing patches.

- **Linux systems:**

Stop all Teamcenter services except FMS services before you begin the update process.

3. Launch TEM:

- **Windows systems:**

Start→Programs→Teamcenter 14→Environment Manager

- **Linux systems:**

```
TC_ROOT/install/tem.sh
```

If you **patched TEM**, make sure you launch the patched TEM.

4. In the **Maintenance** panel, select **Updates Manager** and click **Next**.
5. In the **Apply Updates** panel, enter the following values, and then click **Next**.
 - a. **Update kit location**
Enter the location of the patch files you expanded in step 1.
 - b. **Backup directory**
Enter the location in which you want to store backups of files replaced during the update process.
6. If the patch contains enhancements to features in your configuration, TEM displays the **Optional Configuration Enhancements** panel. Review the list of enhancements and decide whether to install the enhancements:
 - a. Click **View Enhancement Info** for each feature to view information about the enhancements. Note each feature that has enhancements.
 - b. If you want to install the optional feature enhancements, select **Yes**. Otherwise, select **No**.

Warning:

If you select **Yes**, features containing data model objects may be upgraded. That is, database model changes may occur. Siemens Digital Industries Software recommends you back up your database before continuing.

If you select **No**, features containing data model objects are not updated. If you want to perform this update at a later time, you must repeat the patch installation procedures to reinstall the patch, and choose **Yes** to include the template update.
7. Click **Next**.
8. If any feature enhancements require the Teamcenter administrative user password, TEM displays the **Teamcenter Administrative User** panel. Type the user name and password, and then click **Next**.
9. In the **Confirmation** panel, click **Start** to begin patching the server.

Make sure all Teamcenter clients and processes on the host are stopped before you continue. If Teamcenter services or processes are running, the update may fail because TEM cannot copy replacement files from the patch to the installed location. All users logged on to the environment are notified that it will not be available until the update process is complete.

On Windows systems, TEM stops all Teamcenter services during the update process. If you use the .NET web tier, make sure IIS is stopped before you continue.

10. After the server is successfully patched, you must manually update your *TC_DATA* directories.
 - a. Create backups of your current *TC_DATA* directories.
 - b. Expand the *platform\tc\data.zip* file from the temporary location you created in step 1.
 - c. Copy the expanded contents of the **data.zip** file to your *TC_DATA* directories, overwriting existing files.
 - d. Copy the **tc_profilevars.bat** file from the backup to the *TC_DATA* directory.
11. If you have a custom template project created in the Business Modeler IDE client, and you selected **Yes** to install enhancements in step 6, upgrade your custom template project:
 - a. Launch the Business Modeler IDE client.
 - b. Open the custom project.
 - c. Ensure the custom project loads successfully with no errors in the Business Modeler IDE console view.
 - d. Analyze and fix any errors.
 - e. If you made any additional changes to the custom template to fix loading errors, package your custom template.
 - f. Update the *updated-feature* template in the Teamcenter corporate server database:
 - A. Launch TEM in maintenance mode.
 - B. Select the configuration that contains the updated feature.
 - C. Select **Update database (Full Model - System downtime required)**.

TEM displays all templates installed in your database.
 - D. Browse to the location of your custom template package file and select the feature file for the updated template (for example, **feature_custom.xml**).

- E. Confirm your selections, and then click **Start**.

TEM updates the database.

12. Perform any additional steps in the patch **README** file to complete the patch.
13. Restart all Teamcenter-related processes and services (such FSC, database daemons, and so on).
14. Restart the Teamcenter server.

How to update servers in parallel while patching

When patching multiple Teamcenter servers, it is not necessary to wait for a patch operation on one server to complete before you begin patching the next server. Allow the patching of the corporate server to complete, then after this, you can patch the remaining servers concurrently, in parallel.

When you patch your first server, TEM uploads a dataset to the volume that contains key items from the TC_DATA directory. This provides necessary information for patching additional servers. TEM references this dataset during subsequent patching operations, instead of needing an exclusive lock on the Teamcenter database. This allows you to initiate patch operations on your remaining servers at the same time, in parallel. This saves a significant amount of time when updating deployments with multiple servers.

Copy SOA Policy Files

When patching is complete, copy the latest SOA policy files to your patched Teamcenter environment.

Structure Manager and other perspectives that manage structures may show performance degradation after patching, due to missing **BOMLine** properties from SOA policy files. Copying the latest SOA policy files helps prevent this performance degradation.

1. In the directory in which you expanded the patch **.zip** file, open the **tc** directory and locate the **data.zip** file.
2. Expand the **data.zip** file to a temporary directory.
3. Copy *all* files from the resulting **data\soa\policies** directory to the **TC_DATA\soa\policies** directory in your patched Teamcenter environment.

Patching the rich client

Patch the rich client using TEM

Caution:

If the patch contains an updated installer file (**install.zip**), **patch Teamcenter Environment Manager (TEM)** before you begin the following steps.

1. Expand the *product-level_number_platform.zip* file to a temporary location.
2. Shut down services:
 - **Windows systems:**

If you use the .NET web tier, shut down Microsoft Internet Information Services (IIS) before you begin installing patches.
 - **Linux systems:**

Stop all Teamcenter services except FMS services before you begin the update process.
3. Launch TEM:
 - **Windows systems:**

Start→Programs→Teamcenter 14→Environment Manager
 - **Linux systems:**

`TC_ROOT/install/tem.sh`

If you **patched TEM**, make sure you launch the patched TEM.
4. In the **Maintenance** panel, select **Updates Manager** and click **Next**.
5. In the **Apply Updates** panel, enter the following values, and then click **Next**.
 - a. **Update kit location**

Enter the location of the patch files you expanded in step 1.
 - b. **Backup directory**

Enter the location in which you want to store backups of files replaced during the update process.

6. If the patch contains enhancements to features in your configuration, TEM displays the **Optional Configuration Enhancements** panel. Review the list of enhancements and decide whether to install the enhancements:
 - a. Click **View Enhancement Info** for each feature to view information about the enhancements. Note each feature that has enhancements.
 - b. If you want to install the optional feature enhancements, select **Yes**. Otherwise, select **No**.

Warning:

If you select **Yes**, features containing data model objects may be upgraded. That is, database model changes may occur. Siemens Digital Industries Software recommends you back up your database before continuing.

If you select **No**, features containing data model objects are not updated. If you want to perform this update at a later time, you must repeat the patch installation procedures to reinstall the patch, and choose **Yes** to include the template update.

7. Click **Next**.
8. If any feature enhancements require the Teamcenter administrative user password, TEM displays the **Teamcenter Administrative User** panel. Type the user name and password, and then click **Next**.
9. In the **Confirmation** panel, click **Start** to begin patching the rich client.

Make sure all Teamcenter clients and processes on the host are stopped before you continue. If Teamcenter services or processes are running, the update may fail because TEM cannot copy replacement files from the patch to the installed location. All users logged on to the environment are notified that it will not be available until the update process is complete.

On Windows systems, TEM stops all Teamcenter services during the update process. If you use the .NET web tier, make sure IIS is stopped before you continue.

10. Perform any additional steps in the patch **README** file to complete the patch.
11. Restart all Teamcenter-related processes and services (such FSC, database daemons, and so on).
12. Restart the Teamcenter server.

Patch the rich client silently

Teamcenter Environment Manager allows you to install patches silently, with no prompts or user interface:

1. **Patch Teamcenter Environment Manager.**
2. Expand the *patch-id.zip* file to a directory on your local host.
3. Open a command prompt.
4. Change to the *TC_ROOT\install* directory (on Windows systems) or the *TC_ROOT/install* directory (on Linux systems).
5. Type the **tem.bat** command (Windows systems) or **tem.sh** command (Linux systems):

```
tem.bat|sh -patch="patch-location" -pass=password@configID
```

Replace *patch-location* with the full path in which you expanded the *patch-id.zip* file. Replace *password* with the Teamcenter administrator password, and *configID* with the ID of the Teamcenter configuration you are patching.¹

This procedure launches TEM and applies the patch with no user interaction required.

For more information about command line arguments for the **tem.bat/sh** utility, see the *Utilities Reference*.

Note:

If the path to the patch location contains spaces, you must enclose the path in quotation marks ("). For example:

Windows systems:

```
tem.bat -patch="..\Teamcenter patches"
```

Linux systems:

```
tem.sh -patch="/../Teamcenter patches"
```

Patch the web tier

1. Unzip the downloaded patch file (*product-level_number_platform.zip*) to a temporary location. This location is referenced in this procedure as *WEB_FILES_LOC*.

¹ You create the configuration ID when you create the Teamcenter configuration in TEM.

2. Change to the **Web_tier** directory under *WEB_FILES_LOC* and inspect its contents.
3. Locate your existing *WEB_ROOT* directory, which contains the Web Application Manager (**insweb**).
4. Update the Web Application Manager:
 - a. Change to the **Web_tier** directory in the Teamcenter 14.3 software kit.
 - b. Expand the **INSTALL_TCWEB.EXE** file (on Windows systems) or **INSTALL_TCWEB.TZ** file (on UNIX systems) into your existing *WEB_ROOT* directory, overwriting existing files.
5. Open the *WEB_ROOT* directory and run the Web Application Manager (**insweb.bat** on Windows systems or **insweb** on Linux systems).
6. Click **Copy ICDs**.
7. Browse to the **icd** directory and then click **OK**:

Windows systems:

WEB_FILES_LOC\Web_tier\icd

Linux systems:

WEB_FILES_LOC\Web_tier\icd

8. Select the web application in the list corresponding to your web tier installation and click **Modify**.
9. In the **Modify Web Application** dialog box, click **Modify Disk Locations**.
10. In the **Modify Disk Locations** dialog box, click the **Add** button to add the *WEB_FILES_LOC\Web_tier* directory to the **Disk Locations for Install Images** list.
11. In the **Modify Disk Locations** dialog box, click **OK**.
12. In the **Modify Web Application** dialog box, click **Reinstall Solutions**.
13. In the confirmation dialog box, click **Yes** to confirm the changes to the disk location list prior to opening to the **Reinstall Solutions** dialog box.
14. In the **Reinstall Solutions** dialog box, click **OK**.
15. If the **Modify Required Context Parameters** dialog box appears, type the appropriate values for any required context parameters and click **OK**.

16. The Web Application Manager regenerates the web tier web application with the patched files. Make sure a deployable file (WAR) is generated during this process. If it is not, click **Generate Deployable File** in the **Modify Web Application** dialog box.
17. After the deployable file is generated, go to the web tier web application's staging directory and find the deployable file (WAR) under the deployment directory.
18. Take the new deployable file and deploy it to your web application server, replacing the previous deployment. The new deployable file contains the web tier patches.

Review the README file

In the patch download location on Support Center, locate the **Tc14.3.0_patch_patch-number_README.pdf** file, which contains information about the patch. Review this document for details relevant to your environment, and identify any possible additional steps required to complete the patch installation.

Part VI: Appendices

Supplemental procedures and references for installing Teamcenter and Active Workspace.

11. Troubleshooting

Using the upgrade log viewer

At the end of the upgrade process, the **Upgrade Features** panel in TEM displays the results of each step in the upgrade process. Clicking the **Navigate Logs** button in this panel opens the upgrade log viewer.

The upgrade log viewer is a dynamic interface that includes:

- Results of each step in the upgrade process.
- Links to log files for each step.
- Detailed contents of log files.

Use the upgrade log viewer to diagnose and address upgrade failures:

1. In the **Pre-Upgrade** and **Upgrade** sections in the top pane, find the description of a step that failed during the upgrade process.
2. In the **Navigate Logs** pane, locate the step in the list, with the names of the logs generated by that step.
3. Click a link to one of the logs.

The upgrade log viewer displays the contents of the selected log in the **Log Details** pane.

4. Review the log contents to find details about the step that failed and the steps that preceded it.

Finding help for resolving upgrade failures

Depending on the type of error, the log file contents may provide sufficient information to help you resolve the problem, or direct you to other Teamcenter documentation for help. For further help, see the following resources:

- **Teamcenter Upgrade Guide**

Contains troubleshooting information in *Troubleshooting Teamcenter upgrade*.

- **Support Center**

Provides support resources for all Siemens Digital Industries Software products.

Submit information from the upgrade log viewer about specific upgrade failures to Siemens Digital Industries Software support on Support Center:

<https://support.sw.siemens.com>

- **Packaged upgrade logs**

Contains all logs generated during the upgrade process.

When an upgrade fails, TEM automatically packages the upgrade logs into a ZIP archive you can submit to Siemens Digital Industries Software support for review and analysis. The archived upgrade logs are in the following location:

Windows systems:

`TC_ROOT\install\upgrade_logs\install_upgrade_logs_package-ID.zip`

Linux systems:

`TC_ROOT/install/upgrade_logs/install_upgrade_logs_package-ID.zip`

Note:

To access the upgrade log viewer outside TEM, open the following location in a Web browser:

Windows systems:

`TC_ROOT\install\upgrade_display_results.html`

Linux systems:

`TC_ROOT/install/upgrade_display_results.html`

Troubleshooting Microsoft SQL Server

Microsoft SQL Server 2014 performance is poor

If you migrate a database application to Microsoft SQL Server 2014 from a previous version, the database server may consume excessive CPU resources and cause poor performance.

To correct this problem, change the SQL Server 2014 Compatibility Level setting from SQL Server 2014 (120) to SQL Server 2012 (110).

For more information about this issue, see the following Microsoft support article:

<https://msdn.microsoft.com>

Teamcenter update fails with ODBC error

When upgrading a Microsoft SQL Server server, an error similar to the following can occur:

```

+++++
ODBC error. SQLSTATE: 42000 Native error: 5074
Message: [Microsoft][ODBC SQL Server Driver][SQL Server]The column '***'
is
dependent on column '***'.
ODBC error. SQLSTATE: 42000 Native error: 4922
Message: [Microsoft][ODBC SQL Server Driver][SQL Server]ALTER TABLE
ALTER COLUMN
<name> failed because one or more objects access this column.
+++++

```

This error occurs when the upgrade process attempts to modify a column that has a dependent column with an index. Microsoft SQL Server does not allow changes to columns with indexes. Also, local DBA indexes may exist that don't match the standard OOTB template for indexes, so it was not anticipated.

This problem can happen because columns that have manually-created statistics attached cannot have their properties modified without first dropping the statistics object. This to ensure the statistics object accurately reflects the content of the column. Manual creation of statistics objects is important to ensuring query performance if you set `AUTO_CREATE_STATISTICS = OFF`.

An auto-created statistics object does not prevent a modify action to a column because auto-created statistics objects can be removed automatically. But, if the system encounters a manually-created statistics object, it cannot be removed automatically, and may result in an access error.

To resolve this problem, perform the following steps:

1. Delete the index ***.
2. Delete the dependent column ***.
3. Continue the upgrade.
4. Run the **index_verifier** utility to re-create standard OOTB indexes:

```
index_verifier -u=infodba -p= -g=dba -o=DO_IT
```

Troubleshooting Teamcenter upgrade

Possible solutions to upgrade issues

Locate possible solutions to problems you may encounter during Teamcenter upgrade.

Problem	Possible cause	Solution
Errors when uploading or downloading files	FMS_HOME environment variable does not point to the location	Uninstall the rich client as described in the appropriate client installation instructions (for Windows or Linux). Then reinstall the rich client.

Problem	Possible cause	Solution
	of the File Management System (FMS) client executables Incomplete installation of FMS client cache executables	Caution: Always uninstall a rich client using this procedure before installing a new rich client.
	Incompatible configurations of FMS	If two rich clients are installed on the workstation, report the errors to the Teamcenter administrator. When two rich clients are installed on the same workstation, FMS uses the FMS_HOME value set during the first installation. The configuration of the file client cache (FCC) to a file server cache (FSC) must support both deployments, especially if the second deployment is to a different database.
Teamcenter displays an error message when you attempt to access Teamcenter online help	The Teamcenter administrator did not include online help when configuring your rich client	Contact the Teamcenter administrator.
Unable to view or to search Teamcenter online help	The Web browser you are using does not meet requirements to view and search Teamcenter online help	Make sure you are using a Siemens Digital Industries Software-supported Web browser. For information about supported Web browsers, see the Siemens Digital Industries Software Certification Database .
Teamcenter upgrade fails with custom types	A custom data type has the same name as the parent class.	If you have a custom type name with the same name as the parent class, the custom type is identified and added to the custom templates, but the upgrade fails. For example, you have a custom storage_media type that belongs to the storage_media class. Before upgrading Teamcenter, rename the custom type name to anything other than the parent class. For example, rename the storage_media type to storage_media_custom before you upgrade.
Teamcenter upgrade fails with large database issues	The database is not sized properly.	If your upgrade changes or deletes a large number of records, it can fail if the database is not sized properly. Oracle displays an error similar to the following: ORA-01652: unable to extend temp segment by 128 in tablespace TEMP The following upgrade examples can lead to this error: <pre>install -mod_attr infodba \$ {TC_USER_PASSWD} dba POM_application_object owning_user POM_attr_no_pom_backpointer + install -mod_attr infodba \$ {TC_USER_PASSWD}</pre>

Problem	Possible cause	Solution														
		<pre>dba POM_application_object last_mod_user POM_attr_no_pom_backpointer +</pre> These commands update many rows in the pom_backpointer table. The number of rows updated is approximately three times the size of the ppom_application_object table. You can determine the table size with the following SQL command: <pre>SELECT COUNT(puid) FROM ppom_application_object;</pre> To resolve this issue, you must have a large undo or rollback tablespace and a large TEMP tablespace. Increase the size of both the TEMP and undo tablespaces to at least 5 GB each. 1. Use the following SQL command to determine where the data files reside: <pre>SELECT tablespace_name, file_name FROM dba_data_files;</pre> The results may resemble the following: <table><thead><tr><th>TABSPACE- _NAME</th><th>FILE_NAME</th></tr></thead><tbody><tr><td>SYSTEM</td><td>D:\ORACLE\ORADATA\SYSTEM01.DBF</td></tr><tr><td>UNDOTBS1</td><td>D:\ORACLE\ORADATA\UNDOTBS01.DBF</td></tr><tr><td>SYSAUX</td><td>D:\ORACLE\ORADATA\SYSAUX01.DBF</td></tr><tr><td>IDATA</td><td>D:\ORACLE\ORADATA\IDATA01.DBF</td></tr><tr><td>ILOG</td><td>D:\ORACLE\ORADATA\ILOG01.DBF</td></tr><tr><td>INDX</td><td>D:\ORACLE\ORADATA\INDX01.DBF</td></tr></tbody></table> In this case, all data files are located in the D:\ORACLE\ORADATA directory. 2. Add a data file to the TEMP tablespace: <pre>ALTER TABLESPACE TEMP ADD tempfile D:\ORACLE\ORADATA\temp2.dbf' size 5000M;</pre>	TABSPACE- _NAME	FILE_NAME	SYSTEM	D:\ORACLE\ORADATA\SYSTEM01.DBF	UNDOTBS1	D:\ORACLE\ORADATA\UNDOTBS01.DBF	SYSAUX	D:\ORACLE\ORADATA\SYSAUX01.DBF	IDATA	D:\ORACLE\ORADATA\IDATA01.DBF	ILOG	D:\ORACLE\ORADATA\ILOG01.DBF	INDX	D:\ORACLE\ORADATA\INDX01.DBF
TABSPACE- _NAME	FILE_NAME															
SYSTEM	D:\ORACLE\ORADATA\SYSTEM01.DBF															
UNDOTBS1	D:\ORACLE\ORADATA\UNDOTBS01.DBF															
SYSAUX	D:\ORACLE\ORADATA\SYSAUX01.DBF															
IDATA	D:\ORACLE\ORADATA\IDATA01.DBF															
ILOG	D:\ORACLE\ORADATA\ILOG01.DBF															
INDX	D:\ORACLE\ORADATA\INDX01.DBF															

Problem	Possible cause	Solution
		3. Add a data file to the undo tablespace: <pre>ALTER TABLESPACE UNDOTBS01 ADD datafile 'D:\ORACLE\ORADATA\' undotbs2.dbf size 5000M;</pre> To help ensure the upgrade is successful, make sure the redo logs are sized properly (100M each) and reside on a separate disk from the data. If necessary, resize the redo logs. 1. Set the ORACLE_SID environment variable. 2. Log on to Oracle: <pre>sqlplus /nolog connect system/password as sysdba;</pre> 3. Find the list of the redo log groups and see where the files reside. <pre>SELECT GROUP#, TYPE, STATUS, MEMBER FROM V\$LOGFILE;</pre> The results may look like this: <pre>3 ONLINE STALE E:\ORACLE\ORADATA\DPVPERF\RED003.LOG 2 ONLINE E:\ORACLE\ORADATA\DPVPERF\RED002.LOG 1 ONLINE STALE E:\ORACLE\ORADATA\DPVPERF\RED001.LOG</pre> 4. Create the new redo logs as follows. You can change the drive where you can create these, but redo logs should reside on a separate disk from the data. <pre>ALTER DATABASE ADD LOGFILE GROUP 4 ('E:\ORACLE\ORADATA\DPVPERF\ RED004.LOG') SIZE 100M; ALTER DATABASE ADD LOGFILE GROUP 5 ('E:\ORACLE\ORADATA\DPVPERF\</pre>

Problem	Possible cause	Solution
		<pre> RED005.LOG') SIZE 100M; ALTER DATABASE ADD LOGFILE GROUP 6 ('E:\ORACLE\ORADATA\DPVPERF\ RED006.LOG') SIZE 100M; </pre>
		5. Run the following command to force the log switch: <pre> ALTER SYSTEM SWITCH LOGFILE; </pre>
		6. (Optional) Drop redo log groups 1, 2, and 3: <pre> ALTER DATABASE DROP LOGFILE GROUP 1; ALTER DATABASE DROP LOGFILE GROUP 2; ALTER DATABASE DROP LOGFILE GROUP 3; </pre>
		Note: If the group number is the current group and you cannot drop it, run the command to force the log switch to make group 4, 5, or 6 current.
Rich client startup performance is slow	Wait time for hierarchy tree checked-out icon display.	Set the TC_show_checkedout_icon preference to FALSE. Note: Users and administrators can set the TC_show_checkedout_icon preference to TRUE to enhance usability or to FALSE to enhance rich client startup performance. For more information, see the <i>Environment Variables Reference</i>.

Resolving FSC master diagnostic failure

This FSC master diagnostic test performed during upgrade checks the FMS server cache (FSC) to determine whether the FSC was promoted to a master FSC or demoted to a nonmaster FSC without the Teamcenter configuration being updated in Teamcenter Environment Manager (TEM).

For example, if your FSC was promoted to master, the **configuration.xml** and FMS master files may not be synchronized. To continue with the current upgrade, you must correct this discrepancy.

If you manually promoted an FMS server cache (FSC) to a master FSC, you must update the **configuration.xml** file and add the following section in the **<data>** section. The actual value of the **<masterModel>** setting does not matter; the presence of the **<FSCMasterSettings>** node causes TEM to detect an FSC master.

```
<FSCMasterSettings guid="guid">
    <masterModel value="Simple Model" />
</FSCMasterSettings>
```

If you manually demoted an FSC to a nonmaster, you must perform the following steps:

1. Remove the **<FSCMasterSettings>** section from the **<data>** section of the **configuration.xml** file.
2. Add the following section.

```
<FSCServiceNonMasterSettings guid="guid">
    <masterList value="http://fsc_parent_host:4544(0)" />
</FSCServiceNonMasterSettings>
```

The **<masterList>** value must be the current FSC's parent FSC. The **(0)** at the end indicates it is a priority 0 connection.